

Probability and statistics



UNIT-1

Probability



Probability: Probability means possibility. It is a branch of mathematics that deals with the occurrence of a random event. The value is expressed from zero to one.

Sample Space and Event :

In this class, We discuss Random Experiment Sample Space and Event.

Experiment : An experiment is a physical process or action observed. And the result is noted.

Example: Drop a pen. May the pen break or not? Take a note of pen broke or not. Random Experiment: We know all the possible outcomes in a random experiment. We can not predict the exact outcome in advance.

Example: Toss a coin. Possible outcomes are {H, T} We can not predict which outcome. Not Random Experiment: Boil the water. The water will evaporate.

We know the outcome in advance.

Sample Space: A set of all possible outcomes of a random experiment is called sample space. Examples: Toss a coin {H, T}

Toss two coins {HH, HT, TH, TT}

Roll a dice {1, 2, 3, 4, 5, 6}

Event: Every non-empty subset of a sample space of a random experiment is called an event.

Example: Toss two coins Sample space = {HH, HT, TH, TT} getting two heads is an event = {HH} Getting exactly one head is an event = {HT, TH}

Examples



- Probability
- Examples Coin and Dice

In this class, We discuss Probability Examples Coin and Dice.

Random Experiment: Tossing two unbiased coins is our experiment.

1) Find the probability of getting exactly one head.

To find the probability value, we find the sample space.

Sample space = {HH, HT, TH, TT}

Number of elements in sample space $N(S) = 4$

E = Event = Getting exactly one head $E = \{HT, TH\}$

$N(E)$ = Number of elements in the event. $N(E) = 2$

Probability of the event $E = N(E)/N(S)$

$P(E) = N(E)/N(S)$

$P(E) = 2/4$

$= 1/2 = 0.5$

Examples



- **2) Find the probability of getting atleast one head?**
- For the same random experiment. $N(E) = \{HH, HT, TH\} = 3$ $N(S) = 4$
- $P(E) = N(E)/N(S)$
- $= 3/4 =$
- $= 0.75$
- Random Experiment: Roll an unbiased dice
- 1) Probability of getting 1 or 2 on the dice.
- Sample space $= \{1, 2, 3, 4, 5, 6\} = S$
- $N(S) = 6$
- Event $= \{1, 2\} = E$
- $N(E) = 2$
- $P(E) = N(E)/N(S)$
- $= 2/6$
- Random Experiment: Roll two unbiased dice.
- Sample space $= \{(1,1), (1,2), \dots, (6,6)\} = S$ $N(S) = 36$



- **1) Probability of getting sum value showed on the dice is 10**
- Event = E = sum of values on the dice = 10
- $E = \{(5,5), (6,4), (4,6)\}$
- $N(E) = 3$
- $P(E) = N(E)/N(S)$
- $P(E) = 3/36$
- $P(E) = 1/12$
- Probability of getting sum value perfect square.
- Event = $E = \{(1,3), (3,1), (2,2), (3,6), (6,3), (4,5), (5,4)\}$ $N(E) = 7$ $P(E) = N(E)/N(S) = 7/36$



- **Axioms of Probability:**
- Axiom 1: For any event A , $P(A) \geq 0$.
- Axiom 2: Probability of the sample space S is $P(S) = 1$.
- Axiom 3: If $A_1, A_2, A_3, \dots$ are disjoint events, then $P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$

Introduction



Elementary Theorems on Probability

Probability defines the possibility or likelihood of any outcome of a certain event based on the elementary theorems on probability.

Let's know about the **elementary theorems on probability**.



- **Probability Theorems**

- Different **elementary theorems on probability** help us in understanding and easy solving probability questions:
- **Theorem 1:** The sum of probability of happening and not happening of any given event is always unity, i.e., equals 1.
- $P(E) + P(E') = 1$
- **Theorem 2:** The probability of an impossible event is always equal to 0.
- $P(E) = 0$
- **Theorem 3:** The sure events always have 1 as a probability.
- $P(E) = 1$
- **Theorem 4:** The probability of any event is always between 0 to 1.
- $0 < P(E) < 1$



- **Theorem 5:** For any two events, say M and N, the derivation of the formula of probability can be made by applying the formula of the union of two sets.
- Therefore, the formula of event M and N is as follows:
- $P(M \cup N) = P(M) + P(N) - P(M \cap N)$
- Two mutually exclusive events = $P(L \cup M) = P(L) + P(M)$
- **Bayes' Theorem:** Bayes' theorem or conditional probability theorem defines or states the probability of any event based on conditions of other events.
- The formula for Bayes' theorem based on conditional probability is
- $P(M|N)$ = Occurrence of event M on the basis or condition of event N.
- **Law of Total Probability:** If any experiment has multiple numbers of events, then the sum of all these events will always be equal to 1.
- The formula of the law of total probability is
- $P(A_1) + P(A_2) + P(A_3) + P(A_4) + P(A_5) + \dots + P(A_n) = 1$



- **Solved Questions on Probability**
- What is the probability of getting a prime number while rolling a dice in a game of Ludo?
- **Solution:**
- The probability of getting a prime number = $P(A)$
- Total number of outcomes = 6
- Number of favourable outcomes or desired outcomes = 3
[2, 3, 5]
- Probability = $P(A) = \frac{3}{6}$
- $= \frac{1}{2}$
- **Answer:** The probability of getting a prime number while rolling dice is $\frac{1}{2}$, i.e., 50%.



2. There is a box of candies with 4 chocolate candies, 3 strawberry candies, and 3 raspberry candies. What is the probability of getting chocolate candy?

Solution:

The probability of getting a chocolate candy = $P(A)$

Total number of outcomes = 10

Number of favorable outcomes or desired outcomes = 4

Probability = $P(A)$

$$= 4/10$$

$$= \frac{2}{5}$$

Answer: The probability of getting a chocolate candy is $\frac{2}{5}$.



3. What is the probability of getting a heart card from a standard deck of cards?

Solution:

The probability of getting a heart card = $P(A)$

Total number of outcomes = 52 (total number of cards)

Number of favorable outcomes or desired outcomes = 13
(total number of heart cards in a deck)

Probability = $P(A)$

= $13/52$

= $1/4$

Answer: The probability of getting a heart will be $1/4$, i.e., 25%.

RANDOM VARIABLES



- In probability ,a random variable is a quantity whose value depends on the outcome of the random experiment.

random variable as a function that maps from the sample space of an experiment to the real numbers. Mathematically, Random Variable is expressed as,

$X: S \rightarrow R$

where,

X is Random Variable (It is usually denoted using capital letter)

S is Sample Space

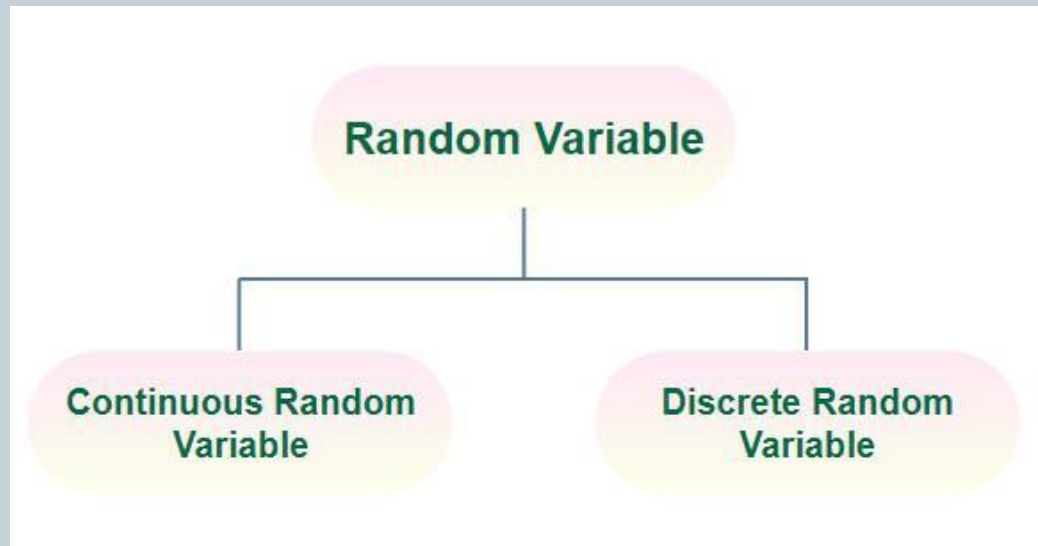
R is Set of Real Numbers

- Examples of random variables:
- 1.The number of heads that appear when a coin is flipped twice.
- 2.The sum of the number on two rolled dice.
- 3.The height of a person.

TYPES OF RANDOM VARIABLES



- **Types of Random Variables**
- Random variables are of two types that are,
- Discrete Random Variable
- Continuous Random Variable



Common probability distribution



- In probability theory and statistics, a probability distribution is a mathematical function that describes the probabilities of occurrence of different possible outcomes for an experiment. It is a mathematical description of a random phenomenon in terms of its sample space and probabilities of events.
- There are two main types of probability distributions:
- **1. Discrete probability distribution**
- **2. Continuous probability distribution.**

Discrete probability distribution



- A discrete probability distribution is one in which the possible values that a random variable can take are countable.
- In continuous probability distribution is one in which the possible value that a random variable can take are uncountable.
- Some examples of probability distributions include
 - **A. Binomial distribution**
 - **B. Poisson distribution**
 - **C. Normal distribution**
- Each distribution has its own unique properties and applications.

BINOMIAL DISTRIBUTION



- **Binomial Distribution in Probability**
- **Binomial Distribution** is a probability distribution used to model the number of successes in a fixed number of independent trials, where each trial has only two possible outcomes: success or failure. This distribution is useful for calculating the probability of a specific number of successes in scenarios like flipping coins, quality control, or survey predictions.



- A **Binomial Distribution** for a random variable $X = 0, 1, 2, \dots, n$ is defined as the probability of success or failure in a series of independent trials. Each trial is independent of the others, and the distribution helps calculate the probability of various outcomes in these trials.



- **Conditions for Binomial Distribution**
- The Binomial distribution can be used in scenarios where the following conditions are satisfied:
- **Fixed Number of Trials:** There are a set number of trials or experiments (denoted by n), such as flipping a coin 10 times.
- **Two Possible Outcomes:** Each trial has only two possible outcomes, often labeled as “success” and “failure.” For example, getting heads or tails in a coin flip.
- **Independent Trials:** The outcome of each trial is independent of the others, meaning the result of one trial does not affect the result of another.
- **Constant Probability:** The probability of success (denoted by p) remains the same for each trial. For example, if you’re flipping a fair coin, the probability of getting heads is always 0.5.



- **Binomial Distribution Formula**
- **The Binomial Distribution Formula** which is used to calculate the probability, for a random variable $X = 0, 1, 2, 3, \dots, n$ is given as
- **$P(X = r) = {}^nC_r p^r q^{n-r}$, $r = 0, 1, 2, 3, \dots$**
- Where,
- **n** = Total number of trials
- **r** = Number of successes
- **p** = Probability of success
- **q** = Probability of failure ($q = 1 - p$)



- **Binomial Distribution Properties**
- Properties of Binomial Distribution are mentioned below:
- There are only two possible outcomes: success or failure, yes or no, true or false.
- There is a finite number of trials given as 'n'.
- The probability of success and failure in each trial is the same.
- Only Success is calculated out of all trials.
- Each trial is independent of any other trial.



- **Binomial Distribution Applications**
- Binomial Distribution is used where we have only two possible outcomes. Let's see some of the areas where Binomial Distribution can be used.
- To find the number of male and female students in an institute.
- To find the likeability of something in Yes or No.
- To find defective or good products manufactured in a factor.
- To find positive and negative reviews on a product.
- Votes are collected in the form of 0 or 1.



- **What is Binomial Distribution Mean and Variance?**
- **Binomial Distribution Mean** tells about the average success obtained in 'n' number of trials. Binomial Distribution Mean is also called Binomial Distribution Expectation. The formula for Binomial Distribution Expectation is given as $\mu = n.p$.
- **Binomial Distribution Variance** is the measurement of the spread of the distribution. The formula of the variance is given by $\sigma^2 = n.p.q$.





- **What is a Binomial Distribution Formula?**
- The Binary Distribution Formula is given as $P(X = r) = {}^nC_r p^r q^{n-r}$. Here $r = 0, 1, 2, 3 \dots$
- Where, p is success, q is failure and is given by
- $q = 1 - p$, and $p, q > 0$ such that $p + q = 1$.

Poisson distribution



- **Binomial Distribution in Probability**
- **Examples**
- **Example 1: A die is thrown 6 times and if getting an even number is a success what is the probability of getting**
 - (i) 4 Successes**
 - (ii) No success**
- **Solution:**
- Given: $n = 6$, $p = 3/6 = 1/2$, and $q = 1 - 1/2 = 1/2$
- $P(X = r) = {}^nC_r p^r q^{n-r}$
- **(i)** $P(X = 4) = {}^6C_4 (1/2)^4 (1/2)^2 = 15/64$
- **(ii)** $P(X = 0) = {}^6C_0 (1/2)^0 (1/2)^6 = 1/64$
-



- **Example 2: A coin is tossed 4 times what is the probability of getting at least 2 heads?**
- **Solution:**
- Given: $n = 4$
Probability of getting head in each trial, $p = 1/2 \Rightarrow q = 1 - 1/2 = 1/2$
- $P(X = r) = {}^4C_r(1/2)^r(1/2)^{4-r}$
 $\Rightarrow P(X = r) = {}^4C_r(1/2)^4$ {Using the laws of Exaponents}
- And we know, Probability of getting at least 2 heads = $P(X \geq 2)$
- $\Rightarrow$ Probability of getting at least 2 heads = $P(X = 2) + P(X = 3) + P(X = 4)$
 $\Rightarrow$ Probability of getting at least 2 heads = ${}^4C_2(1/2)^4 + {}^4C_3(1/2)^4 + {}^4C_4(1/2)^4$
 $\Rightarrow$ Probability of getting at least 2 heads = $({}^4C_2 + {}^4C_3 + {}^4C_4)(1/2)^4$
 $\Rightarrow$ Probability of getting at least 2 heads = $11(1/2)^4 = 11/16$



- **Example 3: A pair of dice is thrown 6 times and getting sum 5 is a success then what is the probability of getting**
- **(i) no success**
- **(ii) two success**
- **(iii) at most two success**
- **Solution:**
- Given: $n = 6$
- 5 can be obtained in 4 ways (1, 4) (4, 1) (2, 3) (3, 2)
- Probability of getting the sum 5 in each trial, $p = 4/36 = 1/9$
Probability of not getting sum 5 = $1 - 1/9 = 8/9$
- (i) Probability of getting no success, $P(X = 0) = {}^6C_0(1/9)^0(8/9)^6 = (8/9)^6$
- (ii) Probability of getting two success, $P(X = 2) = {}^6C_2(1/9)^2(8/9)^4 = 15(8^4/9^6)$
- (iii) Probability of getting at most two successes, $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$
- $\Rightarrow P(X \leq 2) = (8/9)^6 + 6(8^5/9^6) + 15(8^4/9^6)$
-



- **What is a Binomial Distribution Formula?**
- The Binary Distribution Formula is given as $P(X = r) = {}^nC_r p^r q^{n-r}$. Here $r = 0, 1, 2, 3 \dots$
- Where, p is success, q is failure and is given by $q = 1 - p$, and $p, q > 0$ such that $p + q = 1$.
-



Conditions for Binomial Distribution are mentioned below,

- *Only two possible outcomes such as success or failure, yes or no, true or false.*
- *There is a finite number of trials given as 'n'.*
- *Probability of success and failure in each trial is the same.*
- *Only Success is calculated out of all trial.*
- *Each trial is independent of any other trial.*



What is Binomial Distribution Mean and Variance?

Binomial Distribution Mean tells about the average success obtained in 'n' number of trials. Binomial Distribution Mean is also called Binomial Distribution Expectation. The formula for Binomial Distribution Expectation is given as $E[X] = n.p$.

- **Binomial Distribution Variance** is the measurement of the
- spread of the distribution.
- The formula of the variance is given by $\text{Var}[x] = n.p.(1-p)$



What are the Binomial Distribution Characteristics?

- Characteristics of Binomial Distribution are mentioned as follows:
- Possible outcomes such as success or failure, yes or no, true or false.
- Number of observations is finite 'n'.
- Probability of success and failure in each trial is the same.
- Each trial is independent of any other trials.



What is the Purpose of the Binomial Distribution Formula?

Purpose of Binomial Distribution formula is to calculate the probability of obtaining a specific number of successes (often denoted as “ k ”) in a fixed number of independent and identical trials (often denoted as “ n ”) when each trial has only two possible outcomes: success or failure.

POISSON DISTRIBUTION



- The **Poisson distribution** is a type of **discrete probability distribution** that calculates the likelihood of a certain number of events happening in a fixed time or space, assuming the events occur independently and at a constant rate.
- It is characterized by a single parameter, λ (**lambda**), which represents the average rate of occurrence of the event. The distribution is used when the events are rare and the number of occurrences is non-negative and can take on integer values (0, 1, 2, 3,...).



- The key assumptions of the Poisson distribution are:
- 1. Events occur independently of each other.
- 2. The average rate of occurrence (λ) is constant over the given interval.
- 3. The number of events can be any non-negative integer.
-



- **Poisson Distribution Formula**

- Poisson distribution is characterized by a single parameter, lambda (λ), which represents the average rate of occurrence of the events. The probability mass function of the Poisson distribution is given by:
- **$P(X = k) = e^{-\lambda} \lambda^k / k!$**
- Where,
- **$P(X = k)$** is the Probability of Observing k Events
- **e** is the Base of the Natural Logarithm (approximately 2.71828)
- **λ** is the Average Rate of Occurrence of Events
- **k** is the Number of Events that Occur



- **What is Poisson Distribution?**
- Probability distribution which is used to model the number of events that occur in a fixed interval of time or space is called Poisson distribution.
- **When to use Poisson Distribution?**
- Poisson Distribution is generally used to represent those events which are separated over a specific interval of time.
- **What is Poisson Distribution Expected Value?**
- Expected value is the mean of the Poisson Distribution, and is given by the following formula,
- $E[X] = \lambda$
- **What is Lambda in Poisson Distribution?**
- Lambda is the parameter in Poisson Distribution, which is also equal to mean as well as variance.



- **What is the Poisson Distribution Mean and Variance?**
- Mean is the average value whereas variance is the measure of spread for any data including Poisson Distribution.
- **When do we use Poisson Distribution?**
- Poisson distribution is used to model the number of events or occurrences happening in a fixed interval of time or space when these events are rare and random, and the average rate of occurrence is known.
- **Are the Mean and Variance of Poisson Distribution the Same?**
- Yes, in Poisson distribution, mean and variance are equal and have the same value, represented by the parameter λ (lambda).



- **Normal Distribution in Statistics**
- **Normal distribution**, also known as the Gaussian distribution, is a continuous probability distribution that is symmetric about the mean, depicting that data near the mean are more frequent in occurrence than data far from the mean.



- **Key Features of Normal Distribution**
- **Symmetry:** The normal distribution is symmetric around its mean. This means the left side of the distribution mirrors the right side.
- **Mean, Median, and Mode:** In a normal distribution, the mean, median, and mode are all equal and located at the center of the distribution.
- **Bell-shaped Curve:** The curve is bell-shaped, indicating that most of the observations cluster around the central peak, and the probabilities for values further away from the mean taper off equally in both directions.
- **Standard Deviation:** The spread of the distribution is determined by the standard deviation. About 68% of the data falls within one standard deviation of the mean, 95% within two standard deviations, and 99.7% within three standard deviations.



- **Normal Distribution Examples**
- Distribution of Height of People.
- Distribution of Errors in any Measurement.
- Distribution of Blood Pressure of any Patient, etc.



- **Empirical Rule of Standard Deviation**
- ***Empirical Rule states that,***
- 68% of the data approximately fall within one standard deviation of the mean, i.e. it falls between ***{Mean – One Standard Deviation, and Mean + One Standard Deviation}***
- 95% of the data approximately fall within two standard deviations of the mean, i.e. it falls between ***{Mean – Two Standard Deviation, and Mean + Two Standard Deviation}***
- 99.7% of the data approximately fall within a third standard deviation of the mean, i.e. it falls between ***{Mean – Third Standard Deviation, and Mean + Third Standard Deviation}***



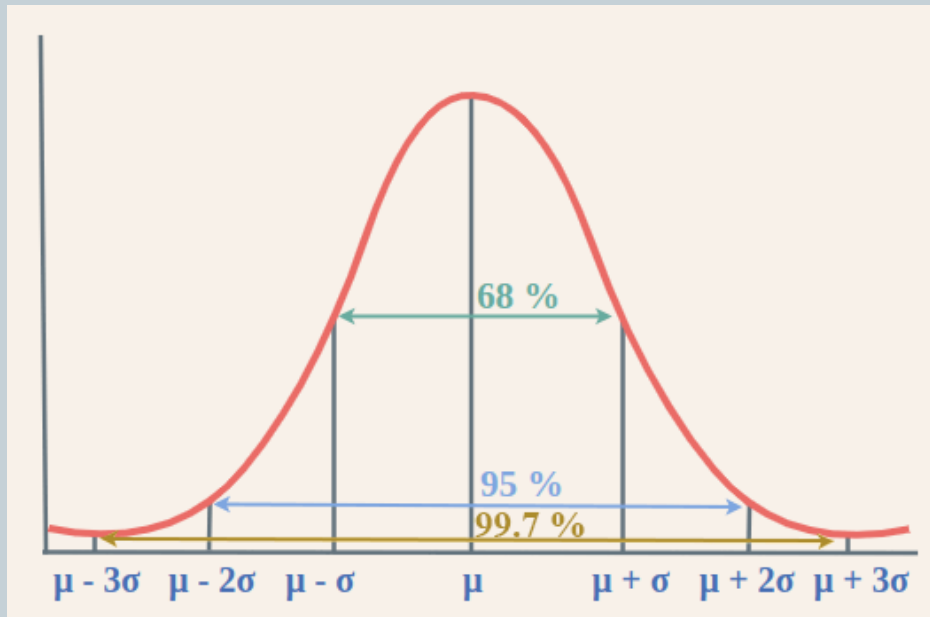
- **Normal Distribution Formula – Probability Density Function (PDF)**
- The formula for the probability density function of Normal Distribution (Gaussian Distribution) is added below:
- where,
- x is Random Variable
- μ is Mean
- σ is Standard Deviation

Probability Density Formula for Normal Distribution

$$f(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$



- **Normal Distribution Graph**
- Studying the graph it is clear that using Empirical Rule we distribute data broadly in three parts. And thus, empirical rule is also called “68 – 95 – 99.7” rule.





- **Sampling distribution**
- A sampling distribution is a probability distribution of a statistic that is obtained from random samples of a specific population.
- When we have a large volume of data, the value of a statistic obtained from these small group of random samples can tell us about the statistic for the entire group.
- **Variability** :It indicates how the outcomes would vary if multiple sample were drawn from the same population.



- Statistics :The study of data gathering ,analysis,interpretation,presentation and organisation is known as statistics.

Sampling distribution procedure

- Select a random sample of a particular size from a given population.
- Compute a desire statistic such as the mean,median or standard deviation for the sample.
- Develop and plot the frequency distribution for each sample statistic computed.
- The resulting graph is known as sampling distribution.



- Sampling distribution of mean
- The sampling distribution of the sample mean is one of the most widely used technique. It focuses on computing the mean of every sample group chosen from the population and plotting all the data points on a graph.

STANDARD ERROR OF PROPORTIONS

In a population, suppose it is possible to make classification into two classes of units as those which 'possess an attribute' and those which 'do not possess the attribute'.

For example -

1. People of a village are classified as 'literate' and 'illiterate'.
2. Students of a college are classified as 'poor' and 'not poor'.
3. Fruits are classified as 'ripe' and 'unripe'.

In a population, let P be the proportion of units which 'possess the attribute'. From such a population, suppose a random sample of size n is drawn. Let x of these n units belong to the class 'possess the attribute'.

Then, $p = x/n$ is the sample proportion of the attribute.

Here, $p = \frac{x}{n}$ has mean $E(p) = P$ and standard error $S.E. (p) = \sqrt{\frac{PQ}{n}}$ where $Q = 1 - P$

Let a random sample of size n_1 be drawn from a population with proportion P_1 of an attribute. Let x_1 units in the sample possess the attribute. Then, the sample proportion is $p_1 = \frac{x_1}{n_1}$.

Let a random sample of size n_2 be drawn from a population with proportion P_2 of the attribute. Let x_2 units in this sample possess the attribute. Then, the sample proportion is $p_2 = \frac{x_2}{n_2}$.

Here, the difference of the sample proportions $(p_1 - p_2)$ has mean $E(p_1 - p_2) = P_1 - P_2$ and standard error $S.E = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$ where $Q_1 = 1 - P_1$ and $Q_2 = 1 - P_2$

Here, if $P_1 = P_2 = P$ the standard error is $\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

Thus, the following are the means and standard errors of some statistics.

Statistic	Mean	Standard Error
$\bar{x}$ (sample mean)	μ	$\frac{\sigma}{\sqrt{n}}$
$\bar{x}_1 - \bar{x}_2$ (difference of means)	$\mu_1 - \mu_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
p (sample proportion)	P	$\sqrt{\frac{PQ}{n}}$
$p_1 - p_2$ (difference of proportions)	$P_1 - P_2$	$\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$
$p_1 - p_2$ (when $P_1 = P_2 = P$)	0	$\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

STATISTICAL INFERENCE

Statistical inference is that branch of statistics which deals with the theory and techniques of making decisions regarding the statistical nature of the population using samples drawn from the population.

Statistical inference has two branches. They are
(i) Theory of estimation and (ii) Testing of hypotheses.

THEORY OF ESTIMATION

Estimator: Estimator of an unknown parameter is a Statistic which specifies the likely value of that parameter.

Estimate: Estimate is a specific value of the estimator for a specified sample. Estimator is a statistic, whereas, estimate is a numerical value.

Example: Suppose a horticulture research centre has evolved a hybrid variety of banana. The centre wants to know the mean weight (μ) of this variety of banana. For this purpose, some bananas are randomly chosen and their mean weight is calculated. This mean weight (sample mean) is proposed as a likely mean of the population. Thus, the sample mean ($\bar{x}$) is an estimator of the population mean (μ). For a specified sample, suppose the sample mean weight is 84 gms., this specific value $\bar{x}$ of is called Estimate of population mean.

There are two types of estimation. They are -

- (i) Point Estimation and
- (ii) Interval Estimation

POINT ESTIMATION:

While estimating an unknown parameter if a single value is proposed as the estimate, such estimation is called Point Estimation.

Thus, based on the sample mean $\bar{x} = 84$ gms., if we conclude that population mean is 84 gms., it is point estimation. Here, $\bar{x}$ is a point estimator of the population mean μ . The specific value 84 gms. is a point estimate of μ . Usually, a point estimator of μ is denoted by $\hat{\mu}$. And so, $\hat{\mu} = \bar{x}$

INTERVAL ESTIMATION

In point estimation, we propose a single value as the estimate of the unknown parameter. In most of the situations, the value so proposed is unlikely to be the actual value of the parameter. Instead, if we propose a small interval around the point estimate as the likely interval to contain the parameter, our proposition would be stronger. This interval which is likely to contain the parameter is called interval estimate.

Interval estimation the estimation of a parameter of a population by computing an interval, or range of values, within which the parameter is most likely to be located is called interval estimation.

Confidence interval: In interval estimation, an interval (T_1, T_2) which is likely to contain the parameter is proposed as estimator of the parameter. The interval (T_1, T_2) is called confidence interval.

Confidence coefficient: The probability that the confidence interval contains the parameter is called confidence coefficient.

Confidence limits: The limits T_1 and T_2 of the confidence interval are called confidence limits.

Note:

- confidence coefficient it is denoted by $(1 - \alpha)$ The confidence coefficient when expressed as percentage is $(1 - \alpha)\%$. In interval estimation, we may write down confidence intervals of different confidence coefficients, say, 95%, 99%, etc
- Confidence interval population mean μ is $(\bar{x} \pm Z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}})$
- 95% confidence interval for μ is $(\bar{x} - 1.96 \times \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \times \frac{\sigma}{\sqrt{n}})$.
It is $(84 - 1.96 \times \frac{5}{\sqrt{100}}, 84 + 1.96 \times \frac{5}{\sqrt{100}})$. It is (83.02, 84.98). And so, with 95% confidence we say that bananas on an average weigh between 83.02 and 84.98 gms.
- 99% confidence interval for μ is $(\bar{x} - 2.58 \times \frac{\sigma}{\sqrt{n}}, \bar{x} + 2.58 \times \frac{\sigma}{\sqrt{n}})$.
- It is $(84 - 2.58 \times \frac{5}{\sqrt{100}}, 84 + 2.58 \times \frac{5}{\sqrt{100}})$ It is (81.61, 85.29). And so, with 99% confidence we say that bananas on an average weight between 81.61 and 85.29 gms.

TESTING OF HYPOTHESES

Testing of hypotheses deals with the verification of validity of presumptions regarding the parameters of the population using samples drawn from the population.

Example: Suppose we presume that 'Mean income of a Bangalorian is Rs.18. per day. In order to test this presumption, some Bangalorians are randomly selected and their mean daily income is found. If this sample mean is in the close neighborhood of Rs. 18, we conclude that 'mean income of a Bangalorian is Rs. 18 per day'. On the other hand, if the sample mean deviates much from Rs. 18, we conclude that 'mean income of a Bangalorian differs from Rs.18'. Thus, if the sample mean is Rs. 18.10. since this value is close to Rs. 18, we conclude that 'mean income of a Bangalorian is Rs. 18 per day'. On the other hand, if the sample mean is Rs.19.60, since this value is far away from Rs. 18, we conclude that 'mean income of a Bangalorian differs from Rs.18'.

STATISTICAL HYPOTHESIS

A statistical hypothesis is an assertion regarding the statistical distribution of the population. It is a statement regarding the parameters of the population. Statistical hypothesis is denoted by H.

Examples

1. H: The population has mean $\mu = 25$.
2. H: The population is normally distributed with mean $\mu = 25$ and standard deviation $\sigma = 2$.

Simple Hypothesis: A hypothesis which completely specifies the statistical distribution of the population is called Simple Hypothesis. It is a hypothesis which specifies all the parameters of the population.

For example,

H: The population is normally distributed with mean $\mu = 25$ and standard deviation $\sigma = 2$ is simple hypothesis.

Composite Hypothesis: A hypothesis which does not completely specify the statistical distribution of the population is called Composite Hypothesis. It is a hypothesis which leaves some parameters unspecified.

For example,

H: the population is normally distributed with mean $\mu = 25$ is a composite hypothesis.

In a test procedure, to start with, a hypothesis is made. The validity of this hypothesis is tested. If the hypothesis is found to be true, it is accepted. On the other hand, if it is found to be untrue, it is rejected.

Null hypothesis: The hypothesis which is being tested for possible rejection is called null hypothesis. The null hypothesis is denoted by H_0 .

Alternative hypothesis: If the null hypothesis is found to be untrue, another hypothesis which contradicts the null hypothesis is accepted. The hypothesis which is accepted when the null hypothesis is rejected is called alternative hypothesis. The alternative hypothesis is denoted by H_1 .

Suppose a null hypothesis is $H_0 : \mu = \text{Rs } 100$ (Mean wage is Rs.100). Depending on the situation, the alternative hypothesis may be any one of the following.

$H_1: \mu \neq \text{Rs.100}$ (Mean wage differs from Rs.100)

$H_2: \mu > \text{Rs. } 100$ (Mean wage is more than Rs.100)

$H_3: \mu < \text{Rs.100}$ (Mean wage is less than Rs.100)

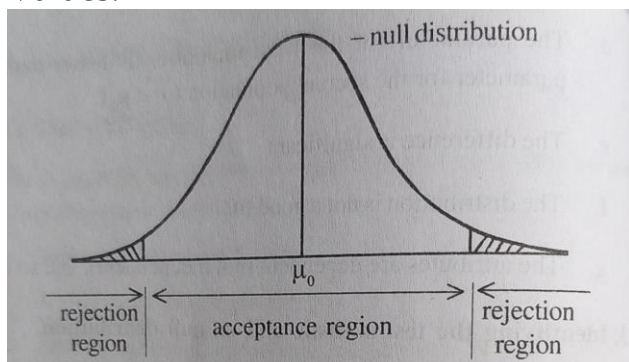
$H_4: \mu = 120$ (Mean wage is 120)

Level of significance: Level of significance is a pre-decided upper limit for the probability of rejection of the null hypothesis when it is actually true. The level of significance is denoted by α . Usually, the pre-decided value α would be 0.05 or 0.01. In other words, it would be 5% or 1%.

Critical Region (Rejection Region): The set of those values of the test statistic which lead of the rejection of the null hypothesis is called critical Region (Rejection Region).

Acceptance region: The set of those values of the test statistic which lead to the acceptance of the null hypothesis is called acceptance region.

Critical Values: The demarcation limits of the critical region are called Critical Values.



ERRORS OF THE FIRST AND THE SECOND KIND (Type 1 and type II errors)

While testing a null hypothesis against an alternative hypothesis one of the following four situations arise.

	Actual fact	Decision based on the sample		Error
1	H_0 is true	Accept H_0	Correct decision	---
2	H_0 is true	Reject H_0	Wrong decision	Type I
3	H_0 is not true	Accept H_0	Wrong decision	Type II
4	H_0 is not true	Reject H_0	Correct decision	---

Here, in situations (2) and (3), wrong decisions are arrived at.

These wrong decisions are termed as Error of the first kind (Type I error) and Error of the second kind (Type II error) respectively. Thus.

(i) Error of the first kind (Type I error) is taking a wrong decision to reject the null hypothesis when it is actually true.

(ii) Error of the second kind (Type II error) is taking a wrong decision to accept the null hypothesis when it is actually not true.

The probability of occurrence of the first kind of error is α . It is the size of the test.

The probability of occurrence of the second kind of error is denoted by β .

The value $(1-\beta)$ is called power of the test. Power of a test is the probability of rejecting H_0 when it is not true.

While testing, the level of significance α is decided in advance. Then, the critical region is determined in such a way that the power $(1-\beta)$ is maximum. Thus, the critical values are based on the level of significance.

TWO-TAILED AND ONE-TAILED TESTS (Two-sided and one-sided tests)

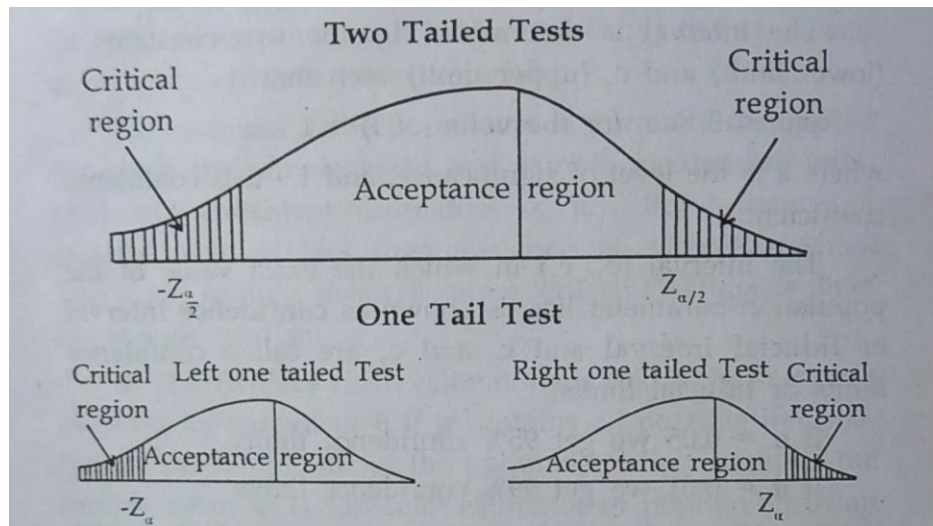
While testing a null hypothesis H_0 against an alternative hypothesis, if the critical region is considered at one tail of the null distribution of the test statistic, the test is one-tailed test (one-sided test).

On the other hand, if the critical region is considered at both the tails of the null distribution of the test statistic, the test is two-tailed test (two-sided test).

or

In testing of H_0 if we consider the critical region of the both tails of the curve of the sampling distribution of a test statistic then the test is called two tailed test.

In testing of H_0 if we consider the critical region of single tail of the curve of the sampling distribution of a statistic then the test is called one-tailed test.



One tail test or two tail test depends on the nature of the alternative hypothesis H_1 . If $H_0: \mu = 1000$ tested against $H_1: \mu > 1000$ is right one tailed test and if $H_0: \mu = 1000$ tested against $H_1: \mu < 1000$ is left one tail test. If $H_0: \mu = 1000$ tested against $H_1: \mu \neq 1000$, is two tailed test.

The following are some of the one-tailed tests.

1. Testing $H_0: \mu = \mu_0$ against $H_1: \mu > \mu_0$
2. Testing $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 < \mu_2$
3. Testing for goodness of fit.
4. Testing for independence of attributes in a contingency table.

The following are some of the two-tailed tests.

1. Testing $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$
2. Testing $H_0: \mu_1 = \mu_2$ against $H_0: \mu_1 \neq \mu_2$
3. Testing $H_0: \sigma = \sigma_0$ against $H_1: \sigma \neq \sigma_0$.

TEST PROCEDURE

The steps in the application of a statistical test procedure for test-ing a null hypothesis are as follows:

1. Setting up the null hypothesis.
2. Setting up the alternative hypothesis.
3. Identifying the test statistic and its null distribution.
4. Identifying the critical region.
5. Drawing a random sample and actually conducting the test.
6. Making the decision (Giving the inference).

1. Setting up the null hypothesis

The hypothesis which is being tested for possible rejection is the null hypothesis. The null hypothesis may be that

- a. the parameter is equal to a given value ($\mu = \mu_0$)
- b. the parameters for the two populations are equal. ($\mu_1 = \mu_2$)
- c. the difference is insignificant (not significant).
- d. The distribution is a good fit.
- e. The attributes are independent, and so on.

2. Setting up the alternative hypothesis.

The hypothesis which is to be accepted when the null hypothesis is rejected is the alternative hypothesis.. It may be that

- a. the parameter is not equal to the given value ($\mu \neq \mu_0$)
- b. the parameter is greater than the given value. ($\mu > \mu_0$)
- c. The parameters for the two populations are not equal ($(\mu_1 \neq \mu_2)$)
- d. The parameter for the first population is lesser than the parameter for the second population ($(\mu_1 < \mu_2)$)
- e. The difference is significant.
- f. The distribution is not a good fit.
- g The attributes are dependent (not independent), and so on.

3. Identifying the test statistic and its null distribution.

The test statistic is the statistic based on whose distribution the test is conducted. The statistical distribution of the test statistic under H_0 is called null distribution.

For testing $H_0: \mu = \mu_0$ (mean is equal to μ_0) the test statistic is $\bar{x}$ the sample mean. Under H_0 the distribution of $\bar{x}$ is $N(\mu_0, \frac{\sigma^2}{n})$. This is the null distribution of $\bar{x}$.

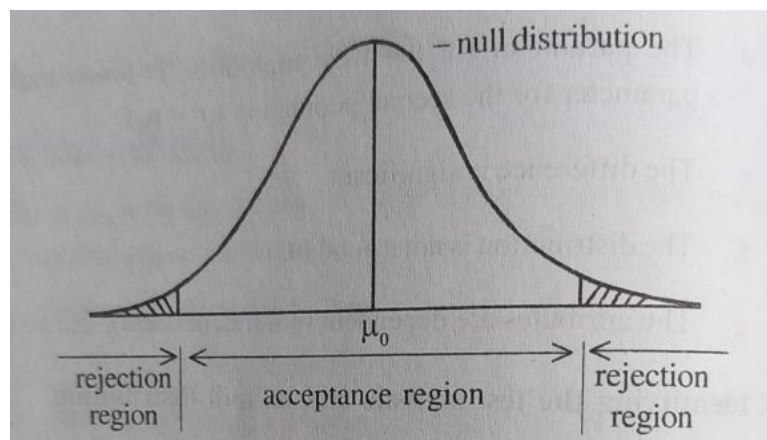
However, for convenience, the standardized form $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ may be considered in which case, the null distribution is $N(0, 1)$

4. Identifying the critical region

The set of those values of the test statistic which lead to the rejection of the null hypothesis is called critical Region (Rejection Region). The set of those values of the test statistic which lead to the acceptance of the null hypothesis is called acceptance region.

The demarcation limits of the critical region are called Critical Values.

The critical region is so formed that when H_0 is true the probability of its rejection is a small pre-decided value. This pre-decided value is called level of significance.



Level of significance is a pre-decided upper limit for the probability of rejection of the null hypothesis when it is actually true. The level of significance is denoted by α . Usually the pre-decided value α would be 0.05 or 0.01. In other words, it would be 5% or 1%.

Depending on the nature of the test and also on the nature of the alternative hypothesis, the critical region may have one part or it may have two parts. If the critical region has one part, the test is called one-sided test (one-tailed test). If the critical region has two parts, the test is called two-sided test (two-tailed test).

5. Drawing a random sample and actually conducting the test.

A random sample of size n is drawn from the population. The value of the test statistic is calculated from the sample values. If this calculated value (observed value) belongs to the critical region, H_0 is rejected in favour of H_1 . On the other hand, if the calculated value belongs to the acceptance region, H_0 is accepted.

6. Making the decision (Giving the inference)

The final decision (inference, conclusion) is announced.

POISSON DISTRIBUTION

A probability distribution which has the following probability mass function (p.m.f.) is called Poisson Distribution.

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots; \lambda > 0 \quad \text{p.m.f.}$$

Here, the variable X is discrete and it is called Poisson variate.

Note 1: λ is the parameter of Poisson distribution. Poisson distribution has only one parameter.

Note 2: A Poisson distribution with parameter λ is denoted by $P(x; \lambda)$ or $P(\lambda)$.

Note 3: Poisson distribution may be treated as a limiting form of binomial distribution under the following conditions.

(Binomial distribution tends to Poisson distribution under the following conditions)

- (i) p is very small ($p \rightarrow 0$)
- (ii) n is very large ($n \rightarrow \infty$) and
- (iii) $np = \lambda$ is fixed

Note 4: Poisson distribution with parameter λ can be written down as follows:

x :	0	1	2	3	
$p(x)$:	$\frac{e^{-\lambda}\lambda^0}{0!}$	$\frac{e^{-\lambda}\lambda^1}{1!}$	$\frac{e^{-\lambda}\lambda^2}{2!}$	$\frac{e^{-\lambda}\lambda^3}{3!}$	

Note 5: The sum of the probabilities is :

$$\begin{aligned}\sum_{x=0}^{\infty} p(x) &= \frac{e^{-\lambda}\lambda^0}{0!} + \frac{e^{-\lambda}\lambda^1}{1!} + \frac{e^{-\lambda}\lambda^2}{2!} + \dots \\ &= e^{-\lambda} \left[\frac{\lambda^0}{0!} + \frac{\lambda^1}{1!} + \frac{\lambda^2}{2!} + \dots \right] \\ &= e^{-\lambda} \cdot e^{\lambda} = e^0 = 1\end{aligned}$$

Note 6: Various theoretical Poisson frequencies are obtained by multiplying the Poisson probabilities by N (Total frequency).

The frequency function is :

$$T_x = N \cdot p(x) = N \cdot \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

EXAMPLES FOR POISSON VARIATE

Many variables which occur in nature vary according to the Poisson law. Some of them are -

1. Number of deaths occurring in a city in a day.
2. Number of road accidents occurring in a city in a day.
3. Number of incoming telephone calls at an exchange in one minute.
4. Number of vehicles crossing a junction in one minute.

- 3 x 2 x 1
3.2!
5. Number of radio-active particles emitted from a radio-active substance in a fraction of a second.
 6. Number of defects in a manufactured article.
 7. Number of deaths of soldiers due to kicks of horses.

MEAN AND VARIANCE OF POISSON DISTRIBUTION

Let X be a Poisson variate with parameter λ .

Then, mean is $E(X) = \lambda$ and Variance is $V(X) = \lambda$

Note : It should be noted that for a Poisson variate Mean and Variance are equal.

CHI-SQUARE TEST OF GOODNESS OF FIT

Suppose there is an observed (empirical) frequency distribution with frequencies, $O_1, O_2, \dots, O_n$. According to certain theoretical assumptions, let a theoretical frequency distribution be fitted to the observed distribution. Let the theoretical frequencies be $E_1, E_2, \dots, E_n$. Suppose we intend to test the null hypothesis –

H_0 : The theoretical frequency distribution is a good fit to the observed frequency distribution.

Against the alternative hypothesis –

H_1 : The theoretical frequency distribution is not a good fit to the observed frequency distribution.

To test H_0 against H_1 , Karl Pearson's Chi-square test of goodness of fit is applied.

Here, the test statistic is –

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = \sum \frac{O_i^2}{E_i} - N$$

where N is the total frequency.

Under H_0 this is a Chi-square variate with $(n - c)$ d.f.

Here, n is the number of terms in the χ^2 and c is the number of constraints.

This test is one-tailed (upper-tailed).

If $\chi^2_{obs} > k_2$, H_0 is rejected. And if $\chi^2_{obs} \leq k_2$, H_0 is accepted. The critical value k_2 is such that $P[\chi^2 > k_2] = \alpha$

For different degrees of freedom and for $\alpha = 0.05$ and $\alpha = 0.01$, the critical values are obtained from the Chi-square table.

The Chi-square test of goodness of fit is applicable subject to the following conditions.

1. The observations should be independent (random).
2. The total frequency N should be large.
3. The theoretical frequencies E_i should be 5 or more. If any E_i is less than 5, it should be pooled with the adjacent frequency (See exercises).
4. If any parameter is estimated from the observed distribution, corresponding to every such estimation, one degree of freedom should be lessened.

HOW TO DECIDE THE DEGREES OF FREEDOM?

Under H_0 , $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$ is Chi-square variate with $(n-c)$ d.f.

Here, n is the number of terms in the χ^2 (after pooling the frequencies which are less than 5 with the adjacent ones).

c is the number of constraints. The theoretical frequencies (E_i) are computed such that $\sum E_i = \sum O_i = N$. This is one constraint. Apart from this, if any parameter is estimated from the observed distribution, every such estimation would be a constraint. Thus, the value of c would be 'one more than the number of parameters estimated from the observed distribution.'

For example, let a χ^2 have $n = 7$ terms. Let 2 parameters be estimated from the observed distribution. Then, $c = 3$. And so, the degrees of freedom is $(n - c) = (7 - 3) = 4$.

χ^2 -distribution – critical values

d.f.	One-tailed test (Lower tail)		One-tailed test (Upper tail)		Two-tailed test		Two-tailed test	
	5%	1%	5%	1%	5%		1%	
	k_1	k_1	k_2	k_2	k_1	k_2	k_1	k_2
1	0.0039	0.0016	3.84	6.63	0.0098	5.02	0.0039	7.88
2	0.10	0.02	5.99	9.21	0.05	7.38	0.01	10.60
3	0.35	0.11	7.81	11.34	0.22	9.35	0.07	12.84
4	0.71	0.30	9.49	13.28	0.48	11.14	0.21	14.86
5	1.15	0.55	11.07	15.09	0.83	12.83	0.41	16.75
6	1.64	0.87	12.59	16.81	1.24	14.45	0.68	18.55
7	2.17	1.24	14.07	18.48	1.69	16.01	0.99	20.28
8	2.73	1.65	15.51	20.09	2.18	17.53	1.34	21.96
9	3.33	2.09	16.92	21.67	2.70	19.02	1.73	23.59
10	3.94	2.56	18.31	23.21	3.25	20.48	2.16	25.19
11	4.57	3.05	19.68	24.72	3.82	21.92	2.60	26.76
12	5.23	3.57	21.03	26.22	4.40	23.34	3.07	28.30
13	5.89	4.11	22.36	27.69	5.01	24.74	3.57	29.82
14	6.57	4.66	23.68	29.14	5.63	26.12	4.07	31.32
15	7.26	5.23	25.00	30.58	6.26	27.49	4.60	32.80
16	7.96	5.81	26.30	32.00	6.91	28.85	5.14	34.27
17	8.67	6.41	27.59	33.41	7.56	30.19	5.70	35.72
18	9.39	7.01	28.87	34.81	8.23	31.53	6.26	37.16
19	10.12	7.63	30.14	36.19	8.91	32.85	6.84	38.58
20	10.85	8.26	31.41	37.57	9.59	34.17	7.43	40.00
21	11.59	8.90	32.67	38.93	10.28	35.48	8.03	41.40
22	12.34	9.54	33.92	40.29	10.98	36.78	8.64	42.80
23	13.09	10.20	35.17	41.64	11.69	38.08	9.26	44.18
24	13.85	10.86	36.42	42.98	12.40	39.36	9.89	45.56
25	14.61	11.52	37.65	44.31	13.12	40.65	10.52	46.93
26	15.38	12.20	38.89	45.64	13.84	41.92	11.16	48.29
27	16.15	12.88	40.11	46.96	14.57	43.19	11.81	49.64
28	16.93	13.56	41.34	48.28	15.31	44.46	12.46	50.99
29	17.71	14.26	42.56	49.59	16.05	45.72	13.12	52.34
30	18.49	14.95	43.77	50.89	16.79	46.98	13.79	53.67
40	26.51	22.16	55.76	63.69	24.43	59.34	20.71	66.77
60	43.19	37.48	79.08	88.38	40.48	83.30	35.53	91.95
80	60.39	53.54	101.88	112.33	57.15	106.63	51.17	116.32
100	77.93	70.06	124.34	135.81	74.22	129.56	67.33	140.17

CHI-SQUARE (χ^2) DISTRIBUTION

A continuous probability distribution which has the following probability density function (p.d.f.) is called Chi-Square distribution.

$$f(\chi^2) = \frac{\left(\frac{1}{2}\right)^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} e^{-\frac{x^2}{2}} (x^2)^{\frac{n}{2}-1}, \quad 0 \leq x^2 < \infty$$

Here, the variable χ^2 is called Chi-square variate with n degrees of freedom (d.f.)

Note 1: Chi-square distribution has one parameter, namely, n which is called degrees of freedom (d.f.)

The degrees of freedom of χ^2 is the number of independent variable components that it has.

Note 2: Chi-square variate is continuous and its range is $(0, \infty)$

Note 3: Chi-square variate with n d.f. is denoted by $\chi^2(n)$

Note 4: $\Gamma\left(\frac{n}{2}\right)$ is read as 'Gamma ($\frac{n}{2}$)' and it is a positive constant for given n

Note 5: Let $Z_1, Z_2, \dots, Z_n$ be n independent standard normal variates.

And let $\chi^2 = Z_1^2 + Z_2^2 + \dots + Z_n^2$

Then, χ^2 is a chi-square variate with n degrees of freedom (d.f.)

APPLICATIONS OF CHI-SQUARE DISTRIBUTION

The chi-square distribution has many uses in the field of testing of hypothesis. Some of them are –

1. to test whether a population has given variance.
2. to test 'goodness of fit' of a theoretical distribution to an observed data.
3. to test independence of attributes in a contingency table.

More details are provided in the chapters ahead.

STUDENT'S t - DISTRIBUTION

A continuous probability distribution which has the following probability density function (p.d.f.) is called Student's t-distribution.

$$f(t) = \frac{1}{\sqrt{n} \beta\left(\frac{1}{2}, \frac{n}{2}\right)} \frac{1}{\left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}}, \quad -\infty < t < \infty$$

Here, the variable t is called student's t -variable with n degrees of freedom (d.f.)

Note 1: t -distribution has one parameter, namely, n which is called degrees of freedom (d.f.).

Note 2: t -distribution has range $(-\infty, \infty)$ and it is symmetrical.

Note 3: t -variate with n d.f. is denoted by $t(n)$

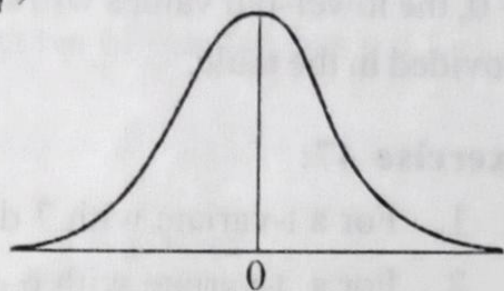
Note 4: $\beta\left(\frac{1}{2}, \frac{n}{2}\right)$ is read as 'beta $\frac{1}{2}, \frac{n}{2}$ ' and it is a positive constant for given n

Note 5: The distribution is introduced in 1908 by W.S. Gosset who published articles in the name 'Student'.

PROPERTIES (FEATURES) OF STUDENT'S t-DISTRIBUTION

A student's t-variate with n d.f. has p.d.f. –

$$f(t) = \frac{1}{\sqrt{n} \beta\left(\frac{1}{2}, \frac{n}{2}\right)} \frac{1}{\left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}}, \quad -\infty < t < \infty$$



1. Its range is $(-\infty, \infty)$
2. It is symmetrical (non-skew) and bell shaped.
3. Its mean is 0. (Median = Mode = Mean = 0)
4. Its variance is $\frac{n}{n-2}$ for $n > 2$ and it is greater than 1.
5. It is leptokurtic.
6. For large n , Student's t-variate tends to Standard Normal Variate.

APPLICATIONS OF t-DISTRIBUTION

Student's t-distribution is very frequently used in tests concerning population means based on small samples. The situations where it is applicable are –

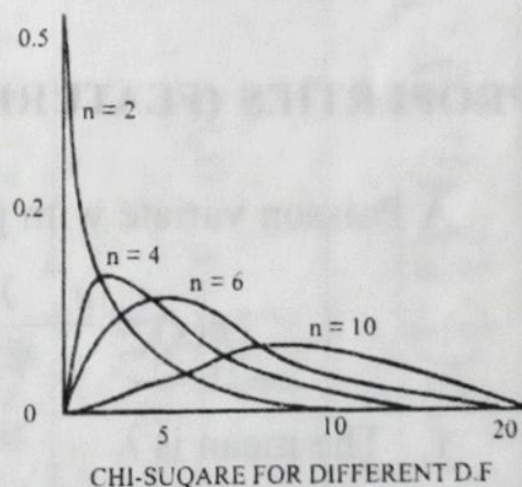
1. to test whether a population has given mean.
2. to test whether the means of two populations are equal.
3. to test whether the means of two populations are equal when the observations are paired.

PROPERTIES (FEATURES) OF CHI-SQUARE DISTRIBUTION

A Chi-square variate with n degrees of freedom has p.d.f. –

$$f(t) = \frac{\left(\frac{1}{2}\right)^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} e^{-\frac{x^2}{2}} (x^2)^{\frac{n}{2}-1}, \quad 0 \leq x^2 < \infty$$

1. Its range is $(0, \infty)$
2. Its mean is n
3. Its variance is $2n$
4. Its mode is $(n - 2)$ for $n \geq 2$.
5. It is positively skew



6. For large n , the variable $Z = [\sqrt{2x^2} - \sqrt{(2n - 1)}]$ approximately has standard normal distribution.

PROPERTIES (FEATURES) OF BERNOULLI DISTRIBUTION

A random variable X which assumes values 1 and 0 with respective probabilities p and $q = 1 - p$ is called Bernoulli variable.

1. The mean is p
2. The variance is pq
3. Let $X_i (i = 1, 2, \dots, n)$ be i.i.d. Bernoulli variates. Then, $X_1 + X_2 + \dots + X_n$ is a binomial variate with parameters n and p .

PROPERTIES (FEATURES) OF BINOMIAL DISTRIBUTION

A Binomial variate with parameters n and p has p.m.f.-

$$p(x) = {}^nC_x p^x q^{n-x}, x = 0, 1, 2, \dots, n$$

1. The mean is np (**Mean > Variance**).
2. The variance is npq . The standard deviation is $\sqrt{npq}$
3. If $p = 0.5$, the distribution is symmetrical
4. Binomial distribution tends to Poisson distribution as $n \rightarrow \infty$, $p \rightarrow 0$ and $\lambda = np$ is fixed
5. Binomial distribution tends to Normal distribution if p as well as q are not very small and $n \rightarrow \infty$

PROPERTIES (FEATURES) OF POISSON DISTRIBUTION

A Poisson variate with parameter λ has p.m.f.-

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$$

1. The mean is λ
2. The variance is λ . The standard deviation is $\sqrt{\lambda}$. **The mean and variance are equal.**
3. Poisson distribution is a limiting form of binomial distribution where $n \rightarrow \infty$, $p \rightarrow 0$ and $\lambda = np$ is fixed
4. Poisson distribution tends to normal distribution as $\lambda \rightarrow \infty$ (for large λ)

DISTRIBUTIONS IN A NUTSHELL

Discrete Distribution	Parameters	Probability function	Range	Mean	Variance
Bernoulli	p	$p^x q^{1-x}$	$x = 0, 1$	p	pq
Binomial	n, p	${}^n C_x p^x q^{n-x}$	$x = 0, 1, 2, \dots, n$	np	npq
Poisson	λ	$\frac{e^{-\lambda} \lambda^x}{x!}$	$x = 0, 1, 2, \dots, \infty$	λ	λ
Continuous distribution					
Normal	μ, σ	$\frac{1}{\sigma\sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right\}$	$-\infty < x < \infty$	μ	σ^2
Chi-square	n	$\frac{\left(\frac{1}{2}\right)^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}\right)} e^{-\frac{x^2}{2}} (x^2)^{\frac{n}{2}-1}$	$0 \leq x^2 < \infty$	n	$2n$
Student's t	n	$\frac{1}{\sqrt{n} \beta\left(\frac{1}{2}, \frac{n}{2}\right)} \frac{1}{\left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}}$	$-\infty < t < \infty$	0	$\frac{n}{n-2}$

Department of Computer Science

QUESTION BANK

Class: BCA II Sem

Sub: Probability & Statistics

Unit I: Probability

2 Marks

1. Define Probability by following approaches.
 - a) Classical approach
 - b) Empirical approach
 - c) Axiomatic approach
2. Define the following terms
 - i) Random Experiment
 - ii) Trial & Event
 - iii) Exhaustive Cases
 - iv) Favourable Cases or Events
 - v) Mutually Exclusive Events or Cases
 - vi) Equally Likely Cases
 - vii) Independent Events
3. Define Conditional probability.
4. Define Random Variable with example.
5. Define Discrete Random Variable & Continuous Random Variable with example.
6. Define Probability Mass Function & Probability Density Function.
7. Give the meaning of Probability Distribution.
8. List Discrete Probability Distributions.
9. List Continuous Probability Distributions.
10. Define Bernoulli variate. Give its PMF.
11. Solve the simple problems based on definitions.

5 Marks

1. State and Prove Addition Theorem of Probability.
2. Write a note on Binomial distribution.
3. Explain Poisson distribution.
4. Discuss in detail Normal Distribution.
5. Discuss in detail Sampling Distribution.
6. Solve problems on Addition Theorem, Conditional Probability Theorem and Baye's Theorem

10 Marks

1. State and Prove Compound Theorem of Probability.
2. State and Prove Baye's Theorem of Probability.
3. Solve Problems on Baye's Theorem.

Unit II: Estimation

2 Marks

1. Define Estimation. Mention its types.
2. Define Point Estimation.
3. Define Interval Estimation.
4. Define Confidence Interval
5. Problems on Confidence Interval for Single Mean and Single Proportion.

5 Marks

1. Explain Interval Estimation for Large Samples.
2. Explain Confidence Limits for Single Mean.
3. Discuss Confidence Limits for Single Proportion.
4. Solve problems on single mean and single proportion.

Unit III: Testing of Hypothesis I

2 Marks

1. Define the following:
 - i) Hypothesis, Simple Hypothesis, & Composite Hypothesis
 - ii) Test of Significance
 - iii) Null Hypothesis & Alternative Hypothesis
 - iv) Type I & Type II Errors
 - v) Critical Region

s

Unit IV: Correlation & Regression

2 Marks

1. Define Correlation. List its types.

2. Define line of Regression.
3. What is Spearman's Rank Correlation coefficient ?

5 Marks

1. Define Correlation. Explain various types of correlation with suitable examples.
2. Write a note on Scatter diagram.
3. Calculate Karl Pearson's coefficient of correlation between expenditure on advertising and sales from the data given below.

Advertising expenses ('000 Rs) : 39 65 62 90 82 75 25 98 36 78
 Sales(lakh Rs.) : 47 53 58 86 62 68 60 91 51 84

4. Write a note on Line of Regression.

10 Marks

1. From the data given below find:
 - a) The two regression coefficients
 - b) The coefficients of correlation between marks in Economics and Statistics

Marks in Economics: 25 28 35 32 31 36 29 38 34 32
 Marks in Statistics: 43 46 49 41 36 32 31 30 33 39

2. Ten competitors in a beauty contest are ranked by three judges in the following order:

1 st Judge	:	1	6	5	10	3	2	4	9	7	8
2 nd Judge	:	3	5	8	4	7	10	2	1	6	9
3 rd Judge	:	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient to determine which pair of judges has the nearest approach to common tastes in beauty.

3. Rankings of 10 trainees at the beginning (x) and at the end (y) of a certain course are given below:

Trainees	:	A	B	C	D	E	F	G	H	I	J
X	:	1	6	3	9	5	2	7	10	8	4
Y	:	6	8	3	7	2	1	5	9	4	10

Calculate Spearman's rank correlation coefficient.

Assignment Questions

1) Calculate Karl Pearson's coefficient of correlation

X	54	60	49	57	68	55	62	60	58
y	3.4	3.2	3.6	4	2.4	3.6	4.1	2.5	3.5

2) Calculate coefficient of rank correlation

X	36	43	47	28	35	50	40
y	73	44	35	30	20	36	40

3) A box has 40 white, 10 red and 8 green marbles. Two marbles are randomly drawn from the box. Find the probability that (i) both the drawn marbles are red (ii) one marble is red and the other is white or green.

4) A box has 100 tickets marked with numbers from 1 to 100. One ticket is drawn at random from the box. Find the probability that the number on the drawn ticket (i) is less than 20 (ii) is even (iii) is a multiple of 5.

5) A can solve 80% of the problems given in a text book. B can solve 70% of the problems in the text. A randomly selected problem from the text is given to A and B. What is the probability that the problem is solved?

6) The probabilities of three persons A, B and C solving a problem are $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$. If all the three independently try to solve, what is the probability that the problem is solved?

7) Find the mean and variance of the following probability distribution

x	1	2	3	4	5	6
P(x)	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

8) A man draws two balls from a bag containing 3 white and 5 black balls. If he is to receive Rs.10 for every white ball that he draws and Rs.7 for every black ball that he draws, find his expectation.

9) The incidence of an occupational disease in an industry is such that the workers have 20% chance of suffering from it. What is the probability that out of six workers 4 or more contract the disease?

10) The probability that a bomb hits a target is $\frac{1}{4}$. Four bombs are aimed at a bridge. Two bomb-hits are enough to destroy the bridge. What is the probability that the bridge is destroyed? What is the probability that all the four bombs hit the bridge?

11) 4 fair coins are tossed 64 times. Write down the theoretical frequencies of the distribution of heads obtained.

12) A survey of 100 families each having 5 children revealed the following distribution:

No of male children	0	1	2	3	4	5	total
No. of families	9	24	35	24	6	2	100

Fit a Binomial distribution.

13) A box has 300 pins. If on an average 1% of pins is defective, find the probability that the box has (i) at least one defective pin (ii) more than 3 defective pins.

14) It is known by experience that 2% articles manufactured by a firm are defective. Articles are supplied in packets of 50 each. (i) What is the probability that a randomly selected packet is free of defective articles? (ii) Among 10,000 packets, how many are expected to be free of defective articles?

15) On an average, 1 in every 50 valves manufactured by a firm is substandard. If valves are supplied in packets of 20 each, in how many of a lot of 1000 packets would you expect substandard valves?

16) For the following data regarding births occurring in a hospital, fit a Poisson distribution.

No. of births per day	0	1	2	3	4	5	≥ 6	total
No. of days	98	110	62	24	6	0	0	300

Activity Questions

1) Calculate Karl Pearson's coefficient of correlation

X y	18-20	20-22	22-24	24-26	26-28
20-23	7	6	1	-	-
23-26	3	8	6	4	8
26-29	1	2	3	8	8
29-32	-	1	1	1	2

2) A bag contains 6 white and 7 black balls. Two balls are drawn at random. Find the probability that they are of (i) the same colour (ii) different colours.

3) Probability of A hitting a target is $\frac{1}{4}$. The probability of B hitting is $\frac{1}{5}$. If both fire once, what is the probability that the target is hit?

4) 2% of the electric bulbs manufactured by a firm are defective. Six bulbs are randomly selected for testing. What is the probability that at least one of the selected bulbs is defective?

5) In an office, 40% of the employees are female. What is the probability that a random selection of 3 employees has females only?

6) The probability of a person winning a game of chess with another person is $\frac{2}{3}$. If he plays three games, what is the probability that he wins at least two of the games?

7) At 10 a.m. there is a city bus service. The number of passengers getting in at the first stop is a Poisson variate with parameter 6. What is the probability that on a particular day nobody gets in at the stop? On how many days of an year would you expect this to happen?

8) To the following data, fit a Poisson distribution and find the expected frequencies.

No. of printing mistakes	0	1	2	3	4	5
No. of pages:	42	33	14	6	4	1

- iii)
- $67 \leq X < 76$
- iv)
- $X < 60$

10) The distribution of monthly income of 3000 workers of a factory confirms Normal law with mean Rs.900 and standard deviation Rs.100. Find (i) the percentage of workers having more than Rs.800 income (ii) the number of workers having less than Rs.600 income.

11) In a city 20% people are non-vegetarians. Find the probability that a random sample consisting of 20 people from the city has at least 2 non-vegetarians. (Use normal approximation)

12) 30% candidates who took the S.S.L.C. examination are girls. Use normal approximation to find the probability that among 50 randomly selected candidates at the most 10 are girls.

13) In a library, half the books are Kannada books. Find the probability that among 20 books selected from the library, at least 5 are Kannada books. (Use normal approximation).

14) Mean and standard deviation of heights of II PUC students are 168 cms, and 3 cms. respectively. Mean and standard deviation of heights of I PUC students are 162 cms. and 4 cms. respectively.

(1) If $\bar{x}_1$ is the mean height of 12 randomly selected II PUC students, write down the mean and standard error of $\bar{x}_1$

(ii) If $\bar{x}_2$ is the mean height of 15 randomly selected I PUC students, write down the mean and standard error of $\bar{x}_2$.

(iii) Also write down the mean and standard error of $(\bar{x}_1 - \bar{x}_2)$

15) Suppose the S.S.L.C. Board results for the state is 33% pass. Find the mean and standard deviation of the proportion p_1 of passes in a school where 36 candidates appeared. If p_2 is the proportion of passes in another school where 47 candidates appeared, find the standard error of p_2 . Also, find the standard error of $(p_1 - p_2)$.

2 marks questions

1. Define null and alternative hypotheses — and give examples.
2. Explain Type I and Type II errors.
3. Discuss the concept of level of significance and its role in hypothesis testing.
4. What is a p-value? How is it interpreted in hypothesis testing?

5 or 10 Marks Question

1. The mean height of a random sample of 100 students is 64 cm and the standard deviation is 5 cm.

Test the statement that the mean height of the population is 67 cm at 5% level of significance

STUDENT UUCMS N0:

**ANJUMAN INSTITUTE OF MANAGEMENT AND COMPUTER APPLICATION, BHATKAL
PREPATORY EXAMINATION JUNE- 2025**

Subject: PROBABILITY AND STATISTICS

Max. Marks: 80

Class: BCA II SEM

Duration: 03 Hours

Part-A

Answer any TEN questions. Each question carries 2 marks

(2 * 10 = 20)

1. What is probability?
2. What is hypothesis testing?
3. What do you mean by regression?
4. What is Type II error?
5. What is correlation?
6. What is mutually exclusive events?
7. What are random variable?
8. What is degree of freedom?
9. Prove that $E(a) = a$, where a is constant.
10. What is interval estimation?
11. Write a formula for Karl's Pearson Co efficient.
12. What is experiment?

Part-B

Answer any FOUR questions. Each question carries 5 marks

(4 * 5 = 20)

1. A card is drawn from a deck of 52 cards . Find the conditional probability of getting a queen or a face card.
2. State and prove Baye's Theorem.
3. The following are the height of 8 mens and their wives . Find the coefficient of correlation between height of husband and wives.

Husband	164	176	178	184	175	167	173	180
Wife	158	164	165	171	163	156	163	169

4. A survey of children revealed that IQ of child (x) and age of mother at the time of giving birth to child(y).

	X	y
Mean	98	28

SD 2 4 , Coefficient of relation $r_{xy} = -0.24$

Estimate the IQ of a child whose mother age is 47 at the time of giving birth of child.

5. A random sample of 100 observation yield sample mean of 150 and variance of 400. Compute 95% and 99% of confidence interval for population mean.
6. A random sample of sample of 50 observation gave a value of SD=24.5. Construct a 95% of confidence interval for population SD.
7. The mean height of random sample of 100 students is 64 and SD is 5. Test the statement that the mean height of population is 67 at 5% of level of significance.
8. A coin is tossed 100 times under identical condition independently yielding 30 heads and 70 tails. Test at 1% level of significance .

Part-C

Answer any 3 questions. Each question carries 10 marks (3* 10 = 30)

- 1 . a) Prove additional theorem of probability.
 b) Prove that i) $\text{Var}(aX) = a^2 \text{Var}(x)$
 ii) $\text{Var}(aX+b) = a^2 \text{Var}(x)$
2. a) The probability distribution of finite variable X is given by following table.

X	-2	-1	0	1	2	3
P(X)	0.1	K	0.2	2k	0.3	k

- i) Find K, Calculate mean and variance.
- j) Find $P[X < 1]$, $P[X > -1]$
 b) What is conditional Probability?
- 3) a) Calculate the co efficient of rank correlation.

X	18	28	35	44	35	26	27	48
Y	83	51	34	34	34	28	46	47

- b) What is critical region?
- 4) a) A random sample of 1000 household in a city revealed that 500 of this had car. Find 95% and 99% of confidence interval proposition of household in city.
 b) what are the properties of good estimator.
- 5) a) An educator claims that an avg IQ of Govt. college students is no more than 110, to test this claims sample of 150 student was taken and given relevant test and their avg IQ score came to 111.2 with SD =7.2 at level of significance 0.01 . Test whether the claim of educator is justified.
 b) What is the probability of having 53 Sundays in a leap year?

1. Compute:

- a) Laspeyres
- b) Paasches
- c) Marshall Edgeworth
- d) Fishers Price index numbers

Commodity	Base year		Current Year	
	Price	Quantity	Price	Quantity
A	6	50	10	60
B	2	100	4	75
C	4	60	6	50
D	10	30	12	40
E	8	40	12	30

Part-D

(Compulsory)

(1 * 10 = 20)

1. Fit a linear trend by the method of least square to the following data and estimate the trend values.

Year	2015	2016	2017	2018	2019	2020	2021
Sales	76	80	130	144	120	135	155

FIRST SEMESTER M.Sc. DEGREE EXAMINATION, APRIL 2023

Computer Science

1.4—PROBABILITY AND STATISTICS

Time : Three Hours

Maximum : 75 Marks

*Answer any five full questions by selecting at least two questions from each part.
All questions carry equal marks.*

Part A

1. (a) What is Statistics ? Describe descriptive and inferential Statistics.
(b) Describe the following terms :
 - (i) Data.
 - (ii) Population.
 - (iii) Census.
2. (a) What is conditional probability ? There are 200 birds in a zoo. 70 birds are male with brown eyes and 100 birds are female with brown eyes 20 of the birds are male with blue eyes and 10 birds are female with blue eyes construct a contingency table.
If a bird is selected at random. What is the probability that the bird is :
 - (i) A female given that it has brown eyes.
 - (ii) A male given that it has blue eyes.
- (b) Define mutually exclusive and not mutually exclusive events. Rahul is deciding which course she wants to take in her next college semester. The probability that she enrolls in a algebra course is 0.30 and the probability she enrolls in a Biology course is 0.70. The probability that she will enroll in a algebra course given that she enrolls in a Biology course is 0.40.
 - (i) What is the probability that she will enroll in an algebra course or a Biology course ?
 - (ii) Are the two events mutually exclusive.
3. (a) State and prove Bayes' theorem.
(b) A multiple choice test contains 20 questions with answer choices A, B, C and D. Only one answer choice to each question represent a correct answer. Find the probability that a student will answer exactly 6 questions correct if he makes random questions in all 20 questions.

Turn over

4. (a) What is continuous and discrete data ? Discuss uniform and normal distributions.
- (b) The test scores of a Physics class with 800 students are normally distributed with a mean of 75 and a standard deviation of 7.
- (i) What percentage of a class has a test score between 68 % and 82 % ?
- (ii) Approximately how many students have a test score greater than or equal 96.
- Solve it by empirical rule.

(7 + 8 = 15 marks)

Part B

5. (a) Define Central Limit Theorem. The entrance exam scores of a certain university has a mean of 74 and a standard deviation of 6.8. The exam scores follows a normal distribution. If a sample of 50 students is selected at random what is the probability that mean exam score of this group is greater than 75.
- (b) What is estimate ? Discuss point estimate and interval estimate.
- (8 + 7 = 15 marks)
6. (a) What is confidence interval ? The average test scores in a Chemistry class is normally distributed with a standard deviation of 6.5 - 100 scores with a sample mean of 82 were selected at random. Find a 90 % confidence interval for the population mean exam score.
- (b) The average test scores of all students in a certain class was measured to be 84 with a standard deviation of 15. If the margin of error was calculated to be 5.36 % at a 95 % confidence level, how many students were in this class ?
- (8 + 7 = 15 marks)
7. (a) Define Students T-distribution. The average weight of 20 students in a certain school was found to be 165 lbs with a standard deviation of 4.5 construct a 95 % confidence interval for the population mean.
- (b) Company XYZ manufactures calculators with an average mass of 450 g. An engineer believes that average weights to be different and decides to calculate the average mass of 50 calculator. State the null and alternative hypothesis.
- (8 + 7 = 15 marks)
8. (a) Define Chi square distribution. Sample of 10 appliances standard deviation of 0.15 V construct a 95 % C.I. for σ^2 and σ .
- (b) What is linear regression ? Write a linear equation that "best fits" the data in the table shown below :

X	1	2	3	4	5	6	7
Y	1.5	3.8	6.7	9.0	11.2	13.6	16

(3 × 5 = 15 marks)

1488 – ISCS(N) – A – 22



FIRST SEMESTER M.SC. (CBCS) DEGREE EXAMINATION, APRIL/MAY 2022

COMPUTER SCIENCE (New)

Paper – 1.4 : Probability and Statistics

Time : 3 Hours]

[Max. Marks : 75

Instructions : Answer any five full questions, choosing at least two questions from each Part. Each full question carry equal marks.

PART – A

1. a) Differentiate between descriptive and inferential statistics.

b) Discuss the following level of measurements

i) Nominal

ii) Ordinal

iii) Interval

iv) Ratio.

(7+8)

2. a) Discuss the following terms with the help of examples

i) Procedure

ii) Event

iii) Sample space.

b) What is a complementary events ? What is complement of rolling a 5 on die ?

(8+7)

[P.T.O.]





3. a) Define conditional probability. There are 500 students in a certain school. 150 students are enrolled in a Algebra course and 80 students are enrolled in a Chemistry course. There are 30 students who are taking both Algebra and Chemistry. If a student is chosen at random. What is the probability that the student is taking chemistry ? Given that the student is also taking Algebra.
- b) What is Bayes' theorem of probability ? Write the formula for Bayes' theorem. (8+7)
4. a) Write the general formula for binomial distribution. What is the binomial distribution if six-sided die is rolled 12 times ? What is the probability of getting a 4 five times ?
- b) What is continuous probability distribution ? Discuss probability density function with the help of example. (8+7)

PART – B

5. a) What is normal distribution ? Discuss the following :
- i) Write the equation of normal probability curve.
 - ii) Show that areas under normal probability curve.
- b) Define Central Limit Theorem. A population mean of 172 lbs Std. Deviation of 10 lbs. Find the probability that a group of 20 men will have an average weight of more than 175 lbs. Assume weight is normally distributed. (7+8)



6. a) What is estimation theory ? Discuss the following :
- i) Point estimation (How a single value is used to approximate a population ?)
 - ii) Confidence interval (Write common confidence level and their critical value).
- b) What is the margin of error ? Write the formula for the margin of error and give the meaning of each notation in the formula. (10+5)
7. a) Discuss the null hypothesis Vs. alternative hypothesis.
- b) Define Chi-square distribution. Write the confidence interval for both variance and standard deviation. (7+8)
8. Write a short note on :
- i) Linear and non-linear correlation.
 - ii) Addition and multiplication rules of probability.
 - iii) The rank correlation. (3×5=15)
-

Definition (2m)

Define probability, sample distribution, dependent-independent, sample space, event, standard error, binomial, poisson, mean, variance, standard deviation, Estimation, pop. parameter, sample statistics, population, correlation, regression, interval estimation, hypothesis testing, Null hypothesis, Alternative hypo, Type I-Type II, Goodness of fit test, Exotic case, classic approach, mutual exclusive events.

Smack

- 1) State & explain condition probability
- 2) State & explain base theorem
- 3) Define random variable & probability distributions & explain its types.
- 4) What is estimation. Explain different types.
- 5) Explain biased & unbiased estimation.
- 6) What is hypothesis testing. explain procedure for hypothesis test.
- 7) List and explain properties of square distribution.
- 8) Explain student-t distribution
- 9) Define degree of freedom give the formula for confidence interval of pop. mean in t-distribution
- 10) List properties of t-distribution.
- 11) Define Critical value & acceptance region
- 12) State the properties of normal distribution
- 13) State & prove addition theorem.
- 14) Write a note on binomial distribution.
- 15) Explain poisson distribution

Define (2m)

pmf, pdf, test of significance, types of correlation, Rank correlation, Spearman's rank correlation, Scatter diagram.

Smash

16) Sampling distribution (explain)

problems

- 1) A man is known to speak lies 1 out of 4 times he throws die & reports that it is a 6. Find the probability that it is actually a 6.
- 2) A random sample of 50 items gives the mean 6.2 & variance 10.24 can it be regarded as drawn from a normal popⁿ with 5.4 & 5% level of significance.
- 3) The probability of 2 mutually exclusive events are $\frac{1}{4}$ & $\frac{1}{6}$ find the probability of occurrence of atleast 1 of these events.
- 4) 2% of fuses manufactured by company are found to be defective find the probability using poisson distribution that a box contains 200 fuses are of
is no defective fuse.
or 3 are more defective fuse.

8 Define Correlation. Explain various types of correlation with suitable ex.

- 5) When a coin is tossed 4 times find the probability of getting
 a) exactly one head
 b) almost 3 heads.
- 6) In a hospital 480 females & 520 male babies were born in a week. Do these figures confirm the hypothesis that male & females are born in equal number given standard error (0.016).

7) The mean ^{yield} of 2 set of plots & their variability are shown below

	Set of 40 plots	Set of 60 plots
mean yield	125.8	124.3
Stand. dev.	34	28

examine whether the difference in variability in yield is significant at 5% level of significance
 $SD = 4.58$

calculate the proportion between two means.

8) The probability distribution of a finite random variable X is given by

X	-2	-1	0	1	2	3
$P(X)$	0.1	k	0.2	$2k$	0.3	k

i) Find k , mean, variance, SD.

evaluate $P(X < 1)$

$P(-1 < X \leq 2)$

9) A random sample of 100 observation yields sample mean $\bar{x} = 150$ and sample variance $S^2 = 400$ compute 95% & 99% confidence interval for popⁿ mean

10) Following information is available in respect of 2 brand of bulbs

Brand 1	Brand A	Brand B
mean life	1300	1248
SD	82	93
sample size	100	100

which brand preferred at 5% level of significance

11) Calcⁿ Kalperson coefficient of correlation b/w expenditure on advertising & sales. from the data given below.

Advertisement expends	39	65	62	90	82	78	25	90	36
Sales	53	58	86	62	68	60	91	78	51
									84

12) From the data given below find 2 regression coefficient & coefficient of correlation b/w marks in economics & statat

Economics	25	28	85	32	31	36	29	38	34	32
Stat	43	46	49	41	36	82	31	30	33	39

13) 10 competitors in beauty contest are ranked by 3 judges in following order

judge 1	1	6	5	10	3	2	4	9	7	8
judge 2	3	5	8	4	7	10	2	1	6	9
judge 3	6	4	9	8	1	2	3	10	5	7

Use rank correlation coefficient to determine which pair of girls is nearest approach to common task in beauty.

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Ranking of 10 trainees at the beginning (x) & at the end (y) of a certain course are given below

Trainees	A	B	C	D	E	F	G	H	I	J
X	1	6	3	9	5	2	7	10	8	4
Y	6	6	3	7	2	1	5	9	4	10

Calc. Spearman rank correlation coefficient.

STUDENT UUCMS N0:

**ANJUMAN INSTITUTE OF MANAGEMENT AND COMPUTER APPLICATION, BHATKAL
PREPATORY EXAMINATION JUNE- 2025**

Subject: PROBABILITY AND STATISTICS

Max. Marks: 80

Class: BCA II SEM

Duration: 03 Hours

Part-A

Answer any TEN questions. Each question carries 2 marks

(2 * 10 = 20)

1. What is probability?
2. What is hypothesis testing?
3. What do you mean by regression?
4. What is Type II error?
5. What is correlation?
6. What is mutually exclusive events?
7. What are random variable?
8. What is degree of freedom?
9. Prove that $E(a) = a$, where a is constant.
10. What is interval estimation?
11. Write a formula for Karl's Pearson Co efficient.
12. What is experiment?

Part-B

Answer any FOUR questions. Each question carries 5 marks

(4 * 5 = 20)

1. A card is drawn from a deck of 52 cards . Find the conditional probability of getting a queen or a face card.
2. State and prove Baye's Theorem.
3. The following are the height of 8 mens and their wives . Find the coefficient of correlation between height of husband and wives.

Husband	164	176	178	184	175	167	173	180
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4. A survey of children revealed that IQ of child (x) and age of mother at the time of giving birth to child(y).

	X	y
Mean	98	28

SD 2 4 , Coefficient of relation $r_{xy} = -0.24$

Estimate the IQ of a child whose mother age is 47 at the time of giving birth of child.

5. A random sample of 100 observation yield sample mean of 150 and variance of 400. Compute 95% and 99% of confidence interval for population mean.
6. A random sample of sample of 50 observation gave a value of SD=24.5. Construct a 95% of confidence interval for population SD.
7. The mean height of random sample of 100 students is 64 and SD is 5. Test the statement that the mean height of population is 67 at 5% of level of significance.
8. A coin is tossed 100 times under identical condition independently yielding 30 heads and 70 tails. Test at 1% level of significance .

Part-C

Answer any 3 questions. Each question carries 10 marks (3* 10 = 30)

- 1 . a) Prove additional theorem of probability.
 b) Prove that i) $\text{Var}(aX) = a^2 \text{Var}(x)$
 ii) $\text{Var}(aX+b) = a^2 \text{Var}(x)$
2. a) The probability distribution of finite variable X is given by following table.

X	-2	-1	0	1	2	3
P(X)	0.1	K	0.2	2k	0.3	k

- i) Find K, Calculate mean and variance.
- j) Find $P[X < 1]$, $P[X > -1]$
 b) What is conditional Probability?
- 3) a) Calculate the co efficient of rank correlation.

X	18	28	35	44	35	26	27	48
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- b) What is critical region?
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1. Compute:

- a) Laspeyres
- b) Paasches
- c) Marshall Edgeworth
- d) Fishers Price index numbers

Commodity	Base year		Current Year	
	Price	Quantity	Price	Quantity
A	6	50	10	60
B	2	100	4	75
C	4	60	6	50
D	10	30	12	40
E	8	40	12	30

Part-D

(Compulsory)

(1 * 10 = 20)

1. Fit a linear trend by the method of least square to the following data and estimate the trend values.

Year	2015	2016	2017	2018	2019	2020	2021
Sales	76	80	130	144	120	135	155

Large Sample Tests

A test of hypothesis is based on the sampling distribution of the test statistic. And so, in order to define the critical region, it is necessary that the sampling distribution be known. Here, the sampling distribution may be one of those which are discussed earlier in this book or it may be one different from these. For small samples ($n < 30$), the test would be based on the respective sampling distribution. However, for large samples ($n \geq 30$), most of the sampling distributions tend to normality, and so, the test may be based on normal distribution. Let us consider some large sample tests which are based on normal distribution.

TEST FOR MEAN

Suppose the mean μ of a population is not known. We want to test whether the mean is a given value μ_0 .

The null hypothesis is –

$$H_0 : \mu = \mu_0 \text{ (Population mean is } \mu_0 \text{)}$$

For a large random sample of size n , under H_0 , the distribution of $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ is $N(0,1)$ and so, the test statistic is $Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

The alternative hypothesis may be any one of the following:

1. $H_1 : \mu \neq \mu_0$. Here, the test is two-tailed.
2. $H_1 : \mu > \mu_0$. Here, the test is one-tailed with critical region at the upper tail.
3. $H_1 : \mu < \mu_0$. Here, the test is one-tailed with critical region at the lower tail.

In the case of two-tailed test, if α is the level of significance, the critical values are $-k_{\alpha/2}$ and $k_{\alpha/2}$. In the case of upper-tailed test, the critical value is k_{α} . In the case of lower-tailed test, it is $-k_{\alpha}$.

For $\alpha = 0.05$, the critical values are $k_{\alpha} = 1.645$ and $k_{\alpha/2} = 1.96$.

For $\alpha = 0.01$, the critical values are $k_{\alpha} = 2.33$ and $k_{\alpha/2} = 2.58$.

Thus, in the case of two-tailed test, if the observed value Z_{obs} for the sample lies outside the interval $(-k_{\alpha/2}, k_{\alpha/2})$, H_0 is rejected. Otherwise, it is accepted. In the case of upper tailed test, H_0 is rejected if $Z_{\text{obs}} > k_{\alpha}$. In the case of lower tailed test, H_0 is rejected if $Z_{\text{obs}} < -k_{\alpha}$.

Note : Here, if σ is not known, the test statistic is $Z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ where s is the sample standard deviation.

TEST FOR EQUALITY OF MEANS

Suppose we intend testing whether means of two populations are equal.

The null hypothesis is -

$H_0 : \mu_1 = \mu_2$ (the means of the two populations are equal).

Under H_0 , let μ be the common mean and let σ_1 and σ_2 be the standard deviations of the two populations.

Let a random sample of size n_1 be drawn from the first population. Let the sample mean be $\bar{x}_1$. Also, let a random sample of size n_2 be drawn from the second population. Let the mean of this sample be $\bar{x}_2$.

Then, under H_0 , the distribution of $Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$ is $N(0,1)$

and it is the test statistic.

The alternative hypothesis may be any one of the following:

1. $H_1 : \mu_1 \neq \mu_2$. Here, the test is two-tailed.
2. $H_1 : \mu_1 > \mu_2$. Here, the test is one-tailed with critical region at the upper tail.
3. $H_1 : \mu_1 < \mu_2$. Here, the test is one-tailed with critical region at the lower tail.

In the case of two-tailed test, if α is the level of significance, the critical values are $-k_{\alpha/2}$ and $k_{\alpha/2}$. In the case of upper tailed test, the critical value is k_{α} . In the case of lower tailed test, it is $-k_{\alpha}$.

Note : Here, if σ_1 and σ_2 are not known, the test statistic is

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad \text{where}$$

s_1 and s_2 are sample standard deviations.

TEST FOR PROPORTION

Suppose the proportion of an attribute in a population is not known. We want to test whether the proportion is a given value P_0 .

The null hypothesis is –

$$H_0 : P = P_0 \text{ (Population proportion is } P_0 \text{).}$$

The alternative hypothesis may be any one of the following:

1. $H_1 : P \neq P_0$. Here, the test is two-tailed.
2. $H_1 : P > P_0$. Here, the test is one-tailed with critical region at the upper tail.
3. $H_1 : P < P_0$. Here, the test is one-tailed with critical region at the lower tail.

In a large random sample of size n from the population, let x units possess the attribute. Then, the sample proportion is $p = \frac{x}{n}$

And so, under H_0 , $Z = \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}}$ is $N(0, 1)$

and it is the test statistic.

In the case of two-tailed test, if α is the level of significance, the critical values are $-k_{\alpha/2}$ and $k_{\alpha/2}$. In the case of upper tailed test, the critical value is k_α . In the case of lower tailed test, it is $-k_\alpha$.

TEST FOR EQUALITY OF PROPORTIONS

Suppose there are two populations with unknown proportions P_1 and P_2 of certain attribute. We wish to test whether the proportions are equal.

The null hypothesis is –

$H_0 : P_1 = P_2$ (Population proportions are equal).

The alternative hypothesis may be any one of the following.

1. $H_1 : P_1 \neq P_2$. Here, the test is two-tailed.
2. $H_1 : P_1 > P_2$. Here, the test is one-tailed with the critical region at the upper tail.
3. $H_1 : P_1 < P_2$. Here, the test is one-tailed with the critical region at the lower tail.

Under H_0 , let P be the common proportion. Let a large random sample of size n_1 be drawn from the first population. Among these n_1 units, let x_1 units possess the attribute so that the sample proportion is $p_1 = \frac{x_1}{n_1}$. Also, let a large random sample of size n_2 be drawn from the second population. Among these n_2 units, let x_2 units possess the

attribute so that the sample proportion is $p_2 = \frac{x_2}{n_2}$

Then, under H_0 , $Z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ is $N(0, 1)$

and it is the test statistic.

Generally, the common proportion P will not be known. And so, it is estimated from the samples.

The estimate is $\hat{P} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$

Thus, the test statistic is $Z = \frac{p_1 - p_2}{\sqrt{\hat{P}\hat{Q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

In the case of two-tailed test, if α is the level of significance, the critical values are $-k_{\alpha/2}$ and $k_{\alpha/2}$. In the case of upper tailed test, the critical value is k_{α} . In the case of lower tailed test, it is $-k_{\alpha}$.

Correlation and Regression

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Correlation - covariation between 2 variables

Correlation is a statistical tool used to measure relationship between two sets of variables in other words

An analysis of the covariation of two or more variables is called correlation

Types of Correlation

① Positive correlation - correlation is positive if the variables vary in the same direction that is if they increase or decrease together $\uparrow\uparrow$ $\downarrow\downarrow$

ex: height and weight of person are +vely correlated.

② Negative correlation - correlation is negative if the variables vary in opposite directions that is if when one variable increases the other variable decreases

ex: production and price of vegetables

③ Perfect correlation - If variables ^{show} exact linear relationship they are said to be perfectly correlated

Perfect correlation may be +ve or -ve

④ Non correlation - If variables do not show related variation they are said to be non-correlated

ex: price of goods & atmospheric temperature.

Causation - A cause and effect relationship b/w two variables is called causation.

ex: production and price of goods
Here an increase in production causes decrease in price
cause - increase in production
effect - decrease in price.

Spurious correlation - The absence of causation is called spurious correlation (Nonsense correlation)

Measurement of correlation

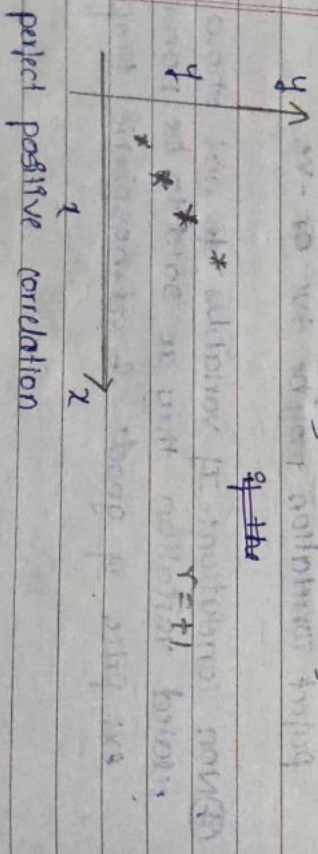
→ ① Scatter diagram

→ ② Karl Pearson's coefficient of correlation

→ ③ Spearman's coefficient of rank correlation.

→ ① Scatter diagram

Scatter diagram is a graphic presentation of bivariate data. Two variables are taken along two axes, and every pair of values in the data is represented by a dot (point) on the graph.
Perfect positive if the points for a line with +ve slope
the variables are truly and perfectly correlated



Perfect negative correlation

If the points for a line with -ve slope the variables are negatively and perfectly correlated.

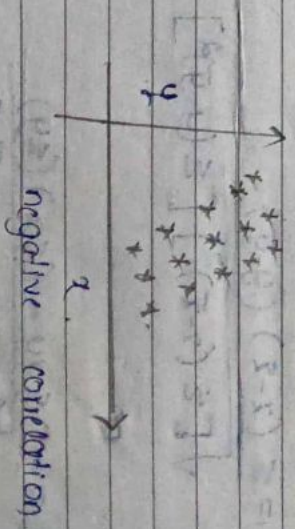


Positive

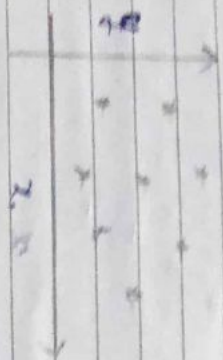
correlation if the points cluster around a line with positive slope. the variables are positively correlated



Negative correlation if the points cluster around a line with negative slope the variables are negatively correlated



Net - Correlation
2 points are spread all over the graph
- variables are non correlated



→ Karl Pearson's coefficient of correlation

$$r = \frac{\text{cov}(x, y)}{\sqrt{v(x)} \sqrt{v(y)}} \quad \text{row-covariations}$$

or

$$r = \frac{\text{cov}(x, y)}{\text{SD}(x) \text{SD}(y)} \quad \text{the variance} = \sigma^2 \quad \text{SD} = \sigma = \sqrt{\sigma^2}$$

standard deviation $r = \frac{\text{cov}(x, y)}{\sigma(x) \sigma(y)}$

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{[\sum (x - \bar{x})^2][\sum (y - \bar{y})^2]}}$$

or

Unpaired

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

$$\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}$$

(unpaired data)

no pairing
only sum data.

Grouped data.

$$r = \frac{N \sum fuv - (\sum fu)(\sum fv)}{\sqrt{[N \sum fu^2 - (\sum fu)^2][N \sum fv^2 - (\sum fv)^2]}}$$

$$\sqrt{[N \sum fu^2 - (\sum fu)^2][N \sum fv^2 - (\sum fv)^2]}$$

Interpretation of coefficient of correlation

1. A positive value of r indicates positive correlation
2. A negative value of r indicates negative correlation
3. If $r = +1$ the correlation is perfect +ve
4. If $r = -1$ the correlation is perfect -ve
5. If $r = 0$ the variables are non correlated

③ Spearman's coefficient of rank correlation

In a bivariate data if the values of variables are rank in increasing or decreasing order the correlation but the ranks is rank correlation it is denoted by $P(Rho)$

If R_1 and R_2 are the ranks of two variables x and y then $d = R_1 - R_2$ is the difference b/w their ranks then coefficient of rank correlation is

$$P = 1 - \left[\frac{6 \sum d^2}{n^3 - n} \right]$$

Rank Correlation

For the computation of rank correlation while ranking the values, tied or more values may be assigned in a such a way, all those values which are equal are assigned with the same average rank and then coefficient of rank correlation is computed.

$$r = 1 - \left[\frac{6 \{ \sum d^2 + CF \}}{n^3 - n} \right]$$

CF = correction

$$CF = \frac{m^3 - m}{12} + \frac{m^3 - m}{12} + 1 \quad d = R_1 - R_2$$

Problems

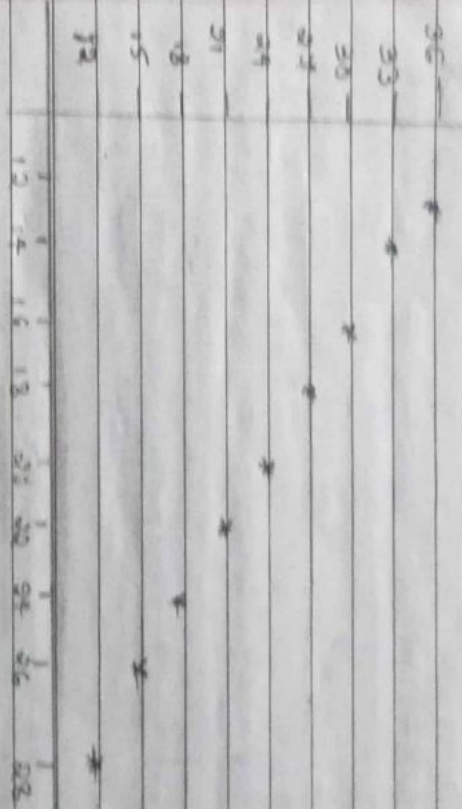
- ① Calculate coefficient of correlation b/w x and y by using scatter diagram

x	y
12	36
14	33
16	30
18	24
20	24
22	21
24	18
26	15
28	12

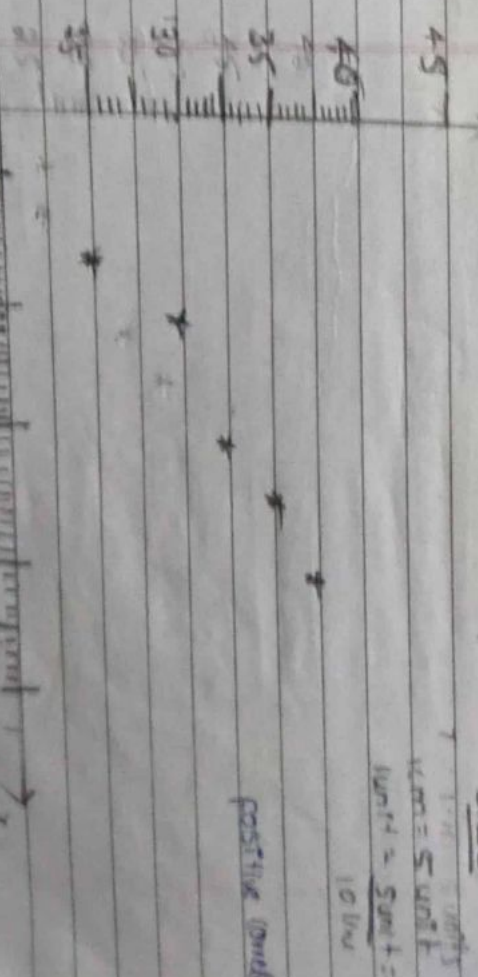
- ② Calculate coefficient of correlation b/w price & supply by scatter diagram

Price (₹)	Supply (kg)
8	10
10	15
15	17
20	20
22	24
24	25

perfect negative correlation



positive correlation



Ungrouped

④

calculate Karl Pearson's coefficient of correlation

x	2	3	4	5	6	7	8	10	14	18
y	4	7	8	9	10	14	18	19	21	25

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 n=7

x	y	x ²	y ²	xy
2	4	4	16	8
3	7	9	49	21
4	8	16	64	32
5	9	25	81	45
6	10	36	100	60
7	14	49	196	98
8	18	64	324	144
10	21	100	441	210
14	25	196	625	350
18	29	324	841	522
Σx	Σy	Σx ²	Σy ²	Σxy
85	140	700	1000	1000

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

$$= \frac{(7 \times 1000) - (85 \times 140)}{\sqrt{[7 \times 700 - (85)^2][7 \times 1000 - (140)^2]}}$$

$$= \frac{2850 - 11950}{\sqrt{(4900 - 7225)(7000 - 19600)}}$$

$$= \frac{406}{\sqrt{196910}} = \frac{406}{443.7326} = 0.961$$

Higher degree positive correlation

by scatter diagram

x	18	20	19	25	21	22	24	26	28	31	34
y	22	24	23	30	25	26	28	31	34		

1 cm = 2 unit
1 cm = 5 unit
10 - 0.5



⑥ Calculate Karl's Pearson's coefficient of correlation

x	y	$u = x - a$	$v = y - b$	u^2	v^2	uv
164	168	0	2	0	4	0
176	174	12	8	144	64	96
178	175	14	9	196	81	126
184	181	20	15	400	225	300
175	173	11	7	121	49	77
167	166	3	0	9	0	0
173	173	9	7	81	49	63
180	179	16	13	256	169	208
		85	61	1207	641	770

deviation: $n \sum uv - (\sum u)(\sum v)$

$$\sqrt{[n \sum u^2 - (\sum u)^2] [n \sum v^2 - (\sum v)^2]}$$

$$= \frac{8 \times 770 - 85 \times 61}{\sqrt{(8 \times 1207) - (85)^2} \sqrt{(8 \times 641) - (61)^2}}$$

$$= \frac{6960 - 5185}{\sqrt{9656 - 7225} \sqrt{5108 - 3721}}$$

$$= \frac{1775}{\sqrt{2431} \sqrt{1407}} = \frac{1775}{\sqrt{3420417}}$$

$$= \frac{1775}{1849436} = 0.959$$

high degree positive correlation r_{xy}

⑦ Calculate Karl's Pearson's coefficient of correlation

x	y	x^2	y^2	xy
4	5	16	25	20
7	8	49	64	56
8	6	64	36	48
5	3	25	9	15
4	5	16	25	20
8	2	64	4	16

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2] [n \sum y^2 - (\sum y)^2]}}$$

$$= \frac{5 \times 154 - 26 \times 27}{\sqrt{(5 \times 154 - (26)^2) (5 \times 159 - (27)^2)}}$$

$$= \frac{765 - 702}{\sqrt{445 - 676} \sqrt{795 - 729}}$$

$$= \frac{63}{\sqrt{94 \times 66}} = \frac{63}{\sqrt{6204}} = \frac{63}{78.765} = 0.799$$

high degree positive correlation r_{xy}

⑧ In bivariate data on x and y variance of x is 15, $V(x) = 49$, $V(y) = 9$, $\text{cov}(x, y) = -17$

find coefficient of correlation b/w x and y

$$r = \frac{\text{cov}(x, y)}{\sqrt{V(x) V(y)}} \text{ formula.}$$

$$r = \frac{-17.5}{\sqrt{49 \times 9}} \Rightarrow \frac{-17.5}{7 \times 3} = \frac{-17.5}{21} = -0.833$$

high degree negative correlation

- 8) In bivariate data $SD(X) = 5$, $SD(Y) = 7$
 $r = 0 - 0.8$ find covariance

$$r = \frac{COV(X, Y)}{SD(X) \cdot SD(Y)}$$

$$-0.8 = \frac{COV(X, Y)}{5 \times 7}$$

$$-0.8 = \frac{COV(X, Y)}{35}$$

$$-0.8 \times 35 = COV(X, Y)$$

$$COV(X, Y) = -0.8 \times 35$$

$$COV(X, Y) = -28$$

- 9) Variables X and Y are perfectly, negatively correlated also $SD(X) = 4.3$ and $SD(Y) = 0.47$

then find covariance of X, Y

$$SD(X) = 4.3 \quad SD(Y) = 0.47$$

$$SD(X) = 0.47 \quad SD(Y) = 4.3$$

$$r = -1 \quad (\text{perfectly negatively correlated})$$

$$r = \frac{COV(X, Y)}{SD(X) \cdot SD(Y)}$$

$$-1 = \frac{COV(X, Y)}{4.3 \times 0.47}$$

$$-1 \times 4.3 \times 0.47 = COV(X, Y)$$

$$COV(X, Y) = -2.021$$

- 10) In bivariate data with 10 observations on X and Y $\sum X = 56$, $\sum Y = 138$, $\sum X^2 = 1357$, $\sum Y^2 = 8136$, $\sum XY = 836$. Find the coefficient of correlation b/w X and Y

$$r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{[n \sum X^2 - (\sum X)^2] [n \sum Y^2 - (\sum Y)^2]}}$$

$$n = 10$$

$$\sum X = 56$$

$$\sum X^2 = 1357$$

$$\sum XY = 836$$

$$\sum Y = 138$$

$$\sum Y^2 = 8136$$

$$(\sum X)^2 = (56)^2 = 3136 \quad (\sum Y)^2 = (138)^2 = 19044$$

$$r = \frac{10 \times 836 - 56 \times 138}{\sqrt{[10 \times 1357 - (56)^2] [10 \times 8136 - (138)^2]}}$$

$$\sqrt{[10 \times 1357 - 3136] [10 \times 8136 - 19044]}$$

$$r = \frac{10 \times 836 - 56 \times 138}{\sqrt{[10 \times 1357 - 3136] [10 \times 8136 - 19044]}}$$

$$\sqrt{[10 \times 1357 - 3136] [10 \times 8136 - 19044]}$$

$$= \frac{8360 - 7728}{\sqrt{[13570 - 3136] [81360 - 19044]}}$$

$$= \frac{632}{\sqrt{[10434 \times 62316]}}$$

$$= 0.32$$

$$\sqrt{10434 \times 62316}$$

$$r = \frac{632}{4915.8055}$$

$$= 0.12856 \approx 0.1286$$

x	0	1	2	3	4	f	fx	fx ²	fy
0	0	0	0	0	0	9	0	0	0
1	4	8	8	0	22	22	22	1	30
2	0	7	12	8	31	62	4	184	142
3	0	3	8	8	24	72	9	216	189
4	0	0	3	5	14	56	16	224	180
Σf	22	33	23	15	100	586	1010	541	
Σfx	22	66	69	60	217				
Σfy	0	22	132	207	240				
Σfxy	0	31	136	186	188	541			

$\Sigma fxy = 541$

$\Sigma fx = 217$
 $\Sigma fy = 541$

grouped data

$r = \frac{N \Sigma fxy - (\Sigma fx)(\Sigma fy)}{\sqrt{[N \Sigma fx^2 - (\Sigma fx)^2][N \Sigma fy^2 - (\Sigma fy)^2]}}$

$= \frac{(100 \times 541) - (217 \times 217)}{\sqrt{[100 \times 586 - (217)^2][100 \times 601 - (217)^2]}}$

$= \frac{54100 - 47089}{\sqrt{[58600 - 47089][60100 - 47089]}}$

$= \frac{8096}{\sqrt{[13656] \times [13011]}}$

$= \frac{8096}{\sqrt{177678216}}$

$= \frac{8096}{13329.5992437} = 0.6073$

$r = 0.6073$

greater than 0.5 +ve

higher degree positive correlation b/w x & y

(12)

X	Y	115	120	125	130
10	-	-	6	11	
20	-	2	4	10	
30	-	3	1	5	
40	3	2	3	1	
50	10	4	5	-	

$$u = \frac{x - \bar{x}}{c}$$

as least value in A
C = width, UB-LW

$$a = 10, b = 115$$

$$c_1 = 10, c_2 = 5$$

$$u = \frac{x - \bar{x}}{c} = \frac{115 - 10}{10} = 10$$

$$v = \frac{y - \bar{y}}{c} = \frac{115 - 10}{10} = 10$$

$$u^2 = 100, v^2 = 100, uv = 100$$

$$u^3 = 1000, v^3 = 1000, uv^2 = 1000, u^2v = 1000$$

$$u^4 = 10000, v^4 = 10000, uv^3 = 10000, u^3v = 10000$$

$$u^5 = 100000, v^5 = 100000, uv^4 = 100000, u^4v = 100000$$

$$u^6 = 1000000, v^6 = 1000000, uv^5 = 1000000, u^5v = 1000000$$

$$u^7 = 10000000, v^7 = 10000000, uv^6 = 10000000, u^6v = 10000000$$

$$u^8 = 100000000, v^8 = 100000000, uv^7 = 100000000, u^7v = 100000000$$

$$u^9 = 1000000000, v^9 = 1000000000, uv^8 = 1000000000, u^8v = 1000000000$$

$$u^{10} = 10000000000, v^{10} = 10000000000, uv^9 = 10000000000, u^9v = 10000000000$$

$$u^{11} = 100000000000, v^{11} = 100000000000, uv^{10} = 100000000000, u^{10}v = 100000000000$$

$$u^{12} = 1000000000000, v^{12} = 1000000000000, uv^{11} = 1000000000000, u^{11}v = 1000000000000$$

$$u^{13} = 10000000000000, v^{13} = 10000000000000, uv^{12} = 10000000000000, u^{12}v = 10000000000000$$

$$u^{14} = 100000000000000, v^{14} = 100000000000000, uv^{13} = 100000000000000, u^{13}v = 100000000000000$$

$$u^{15} = 1000000000000000, v^{15} = 1000000000000000, uv^{14} = 1000000000000000, u^{14}v = 1000000000000000$$

$$① u = \frac{10 - 10}{10} = 0$$

$$② v = \frac{115 - 115}{5} = 0$$

$$- \frac{0}{5} = 0$$

$$③ u = \frac{20 - 10}{10} = \frac{10}{10} = 1$$

$$10 \quad 80$$

$$0 \times 0 \times 0 = 0, 0 \times 1 \times 13 = 0$$

$$0 \times 0 \times 1 = 0$$

$$6 \times 0 \times 2 = 0$$

$$11 \times 0 \times 3 = 0$$

calculation

formula - Karl spearson ungrouped data

$$r = \frac{n \sum uv - (\sum u)(\sum v)}{\sqrt{[n \sum u^2 - (\sum u)^2][n \sum v^2 - (\sum v)^2]}}$$

$$\sqrt{[n \sum u^2 - (\sum u)^2][n \sum v^2 - (\sum v)^2]}$$

$$= (10 \times 169) - (13)^2 \times 130$$

$$\sqrt{[437 - (13)^2][330 - (130)^2]}$$

$$= 11830 - 17810$$

$$= \frac{10 \times [437 - 18769] [330 - 16900] \times 70}{-5980}$$

$$= -5980$$

$$-18332 \times -46570$$

$$-5980$$

$$303701240$$

$$= -5980$$

$$\sqrt{10 \times 437 - 18769} \sqrt{10 \times 330 - 16900}$$

$$= -5980$$

$$\sqrt{[30590 - 18769] [23100 - 16900]} \sqrt{11821 \times 62}$$

$$-5980$$

$$\Rightarrow 8561.94 \Rightarrow -5980 = -5980.698$$

13 Find Karl Pearson coefficient of correlation

X	0-20	20-40	40-60
10-25	10	5	3
25-40	4	40	8
40-55	6	9	15

$a = 17.5 \rightarrow$ least value in X
 $b = 10 \rightarrow$ least value in Y
 $c_1 = 15$
 $c_2 = 20$

X	0-20	20-40	40-60	Y	0-20	20-40	40-60
10-25	10	5	3	18	0	0	0
25-40	4	40	8	52	1	52	56
40-55	6	9	15	30	2	60	78
				100 N		112	134

Y	0-20	20-40	40-60
10-25	10	30	50
25-40	4	40	8
40-55	6	9	15

$u = \frac{x-a}{c}$
 $v = \frac{y-b}{c}$

u	0	1	2
uv	0	1	2
u ²	0	1	4
v ²	0	54	52
uv	0	54	104
uv	0	58	76

$\sum uv = 134$
 $\sum u^2 = 134$
 $\sum v^2 = 106$
 $\sum uv$ should be equal

width of c = U-L-L
 calculation
 $u = \frac{x-a}{c_1}$
 $v = \frac{y-b}{c_2}$

① $\frac{17.5-17.5}{15} = 0$
 ② $\frac{32.5-17.5}{15} = 1$

③ $\frac{47.5-17.5}{15} = 2$
 ④ $\frac{10-10}{80} = 0$
 ⑤ $\frac{30-10}{80} = \frac{20}{80} = \frac{1}{4}$

$\sum uv = 10 \times 0 \times 0 = 0$
 $5 \times 0 \times 1 = 0$
 $3 \times 0 \times 2 = 0$
 $4 \times 1 \times 0 = 0$
 $9 \times 2 \times 1 = 18$
 $15 \times 2 \times 2 = 60$

formula = $r = \frac{N \sum uv - (\sum u)(\sum v)}{\sqrt{[N \sum u^2 - (\sum u)^2][N \sum v^2 - (\sum v)^2]}}$

$r = \frac{100 \times 134 - (112)(134)}{\sqrt{[100 \times 112 - (112)^2][100 \times 106 - (106)^2]}}$

$r = \frac{13400 - 11872}{\sqrt{[17200 - 12544][15800 - 11236]}}$

$= \frac{1528}{\sqrt{4656 \times 4564}}$

$= \frac{1528}{\sqrt{21249984}} = \frac{1528}{4609.7704932}$

$r = 0.331$

there lower degree the correlation b/w x & y

14) For following are the marks of 8 students in stat & maths

marks in stat	marks in maths
25	35
43	47
27	20
35	37
54	63
61	54
37	28
45	40

X	Y	R ₁	R ₂	d = R ₁ - R ₂	d ²
25	35	8	6	2	4
43	47	4	3	1	1
27	20	7	8	-1	1
35	37	6	5	1	1
54	63	2	1	1	1
61	54	1	2	-1	1
37	28	5	7	-2	4
45	40	3	4	-1	1
		Σ d ² = 14			

Total no of observation = 8

value of 1 not repeated

give rank 1, 2, 3, 4, 5, 6, 7, 8 in the same way

$$r = 1 - \left[\frac{6 \sum d^2}{n^3 - n} \right] \rightarrow \text{if values are not repeated}$$

$$r = 1 - \left[\frac{6 (\sum d^2 + c)}{n^3 - n} \right] \rightarrow \text{if values are repeated}$$

In this following table values are not repeated then

$$r = 1 - \left[\frac{6 \sum d^2}{n^3 - n} \right]$$

$$r = 1 - \left[\frac{6 \times 14}{8^3 - 8} \right] = \frac{84}{512 - 8} = \frac{84}{504} = 0.1666$$

$$= 1 - 0.1666$$

$$r = 0.8333$$

higher degree of correlation

15) For the following data ~~structure~~ calculate coefficient of rank correlation

in the following table there is no value repeated in X, and no value repeated in Y

So we formula =

$$r = 1 - \left[\frac{6 \sum d^2}{n^3 - n} \right]$$

X	Y
60	90
110	100
65	80
40	30
70	70
80	80

X	Y	R ₁	R ₂	d = R ₁ - R ₂	d ²
60	90	4	2	2	4
110	100	1	1	0	0
65	80	3	3	0	0
40	30	5	5	0	0
70	70	2	4	-2	4
80	80	6	6	0	0

$\sum d^2 = 8$

$$r = 1 - \left[\frac{6 \sum d^2}{n^3 - n} \right]$$

$n = 6$
total no of obs

$$= 1 - \left[\frac{6 \times 8}{6^3 - 6} \right]$$

higher degree +ve correlation b/w X and Y

$$= 1 - \frac{48}{210}$$

$$\Rightarrow 1 - 0.2285$$

$$\Rightarrow 0.7715$$

16) 6 students are ranked in sports and studies as follows find the coefficient of rank correlation

Student	rank in sports	rank in study
1	2	4
2	3	5
3	4	6
4	5	2
5	6	1
6	1	3

R ₁	R ₂	d = R ₁ - R ₂	d ²
1	4	-3	9
2	5	-3	9
3	6	-3	9
4	2	2	4
5	1	4	16
6	3	3	9

$\sum d^2 = 56$

$$r = 1 - \left[\frac{6 \sum d^2}{n^3 - n} \right]$$

$n = 6$

$$= 1 - \frac{336}{210}$$

$$= 1 - 1.6$$

$$= -0.6$$

higher degree -ve correlation b/w X and Y.

17) In a beauty contest, 10 ladies are ranked by 3 judges as follows,

Judge I	Judge II	Judge III	which two of the 3 judges have closer taste of beauty
4	6	6	
7	5	8	
2	1	2	
1	2	1	
5	9	5	
6	10	9	
7	9	3	
8	6	7	
9	3	4	
10	8	7	

R1	R2	R3	d ₁₂	d ₁₃	d ₂₃	d ₁₂ ²	d ₁₃ ²	d ₂₃ ²
4	6	6	-2	-2	0	4	4	0
7	5	8	2	-1	-3	4	1	9
2	1	2	1	0	-1	1	0	1
1	2	1	-1	0	1	1	0	1
5	9	5	-4	0	4	16	0	16
10	10	9	0	1	1	0	1	1
9	3	10	6	-1	-7	36	1	49
6	7	3	-1	3	4	1	9	16
3	4	4	-1	-1	0	1	1	0
8	8	7	0	1	1	0	1	1

$$f_{12} = \frac{6 \sum d_{12}^2}{n^3 - n} = \frac{6 \times 10}{1000 - 10} = 0.06122$$

$$= 1 - 0.3878$$

$$= 1 - \frac{384}{990}$$

$$f_{12} = 0.6122$$

$$= 1 - 0.3878$$

$$f_{13} = 1 - \frac{6 \sum d_{13}^2}{n^3 - n} = 1 - \frac{6 \times 10}{1000 - 10} = 0.06122$$

$$= 1 - \frac{6 \times 10}{990} = 1 - \frac{564}{990} = 0.4304$$

$$= 1 - \frac{108}{990} = 1 - 0.1090 = 0.891$$

$$f_{13} = 0.891$$

1st and 3rd Judge have closer taste of beauty.

highest value $f_{12} = 0.6122$ highest so they have closest taste of beauty

18) Following are marks in Stat & Maths of 10 students find coefficient of rank correlation

Stat	Maths	Here values are repeated so we formula
43	30	
96	94	
74	84	
38	13	
35	80	
43	18	
82	30	
56	41	
35	48	
80	95	

$$f = 1 - \left[\frac{6 \sum d^2 + (CF)}{n^3 - n} \right]$$

43 repeated 2 times
 35 repeated 2
 30 repeated 2
 18 repeated 2
 10 repeated 2
 9 repeated 2
 8 repeated 2
 7 repeated 2
 6 repeated 2
 5 repeated 2
 4 repeated 2
 3 repeated 2
 2 repeated 2
 1 repeated 2

X	Y	R1	R2	d=R1-R2	d ²
43	30	55	7	-1.5	2.25
96	94	1	2	-1	1
74	84	3	3	0	0
38	13	7	10	-3	9
35	30	8.5	7	1.5	2.25
43	18	5.5	9	-3.5	12.25
22	30	10	7	3	9
56	41	4	5	-1	1
35	48	8.5	4	4.5	20.25
80	95	2	1	1	1
Σd ²					58

n=10

$$f = 1 - \left[\frac{6(\Sigma d^2 + CF)}{n^3 - n} \right]$$

$$= 1 - \left[\frac{6 \times 58 + CF}{10^3 - 10} \right]$$

To find CF

3 values are repeated 43, 35, 30

so take m1, m2, m3 → frequency

m1=43 → repeated 2 times

$$CF = \frac{m_1^3 - m_1}{12} + \frac{m_2^3 - m_2}{12} + \frac{m_3^3 - m_3}{12}$$

Q

m1 = 43 → repeated 2 times ⇒ m1 = 2

m2 = 35 → repeated 2 times ⇒ m2 = 2

m3 = 30 → repeated 2 times ⇒ m3 = 3

$$CF = \frac{m_1^3 - m_1}{12} + \frac{m_2^3 - m_2}{12} + \frac{m_3^3 - m_3}{12}$$

$$= \frac{(2)^3 - 2}{12} + \frac{(2)^3 - 2}{12} + \frac{(3)^3 - 3}{12}$$

$$= \frac{8-2}{12} + \frac{8-2}{12} + \frac{27-3}{12}$$

$$= \frac{6}{12} + \frac{6}{12} + \frac{24}{12}$$

$$= 0.5 + 0.5 + 2 \left[\frac{1}{2} + \frac{1}{2} + \frac{2}{1} \right]$$

$$CF = 3$$

$$f = 1 - \left[\frac{6 \times [58 + 3]}{10^3 - 10} \right]$$

$$= 1 - \left[\frac{6 \times 61}{10^3 - 10} \right]$$

$$= 1 - \left[\frac{366}{1000 - 10} \right]$$

$$= 1 - \frac{366}{990}$$

$$= 1 - 0.3696$$

$$f = 0.6304$$

higher degree of correlation b/w X & Y

19

coefficient of rank correlation

X	Y	R ₁	R ₂	d = R ₁ - R ₂	d ²
18	83	8	1	7	49
28	51	6	2	4	16
(35)	(34)	4.5	6	-1.5	2.25
44	(34)	2	6	-4	16
(35)	(34)	4.5	6	-1.5	2.25
26	28	7	8	-1	1
34	46	3	4	-1	1
48	47	1	3	-2	4

$$\frac{5+6+7+1}{3} = 3$$

$$\sum d^2 = 91.5$$

$$n = 8$$

$$r = 1 - \left[\frac{6 \sum d^2 + (CF)}{n^3 - n} \right]$$

$$= 1 - \left(\frac{6 \times 91.5 + (CF)}{8^3 - 8} \right)$$

CF 6 separated 8 times m1

$$m_1 = 2, m_2 = 3$$

$$CF = \frac{m_1^3 - m_1}{12} + \frac{m_2^3 - m_2}{12}$$

$$= \frac{(2)^3 - 2}{12} + \frac{(3)^3 - 3}{12}$$

$$= \frac{2-2}{12} + \frac{27-3}{12}$$

$$CF = \frac{6}{12} + \frac{24}{12} = \frac{30}{12} = 2.5$$

$$r = 1 - \left[\frac{6 \times (91.5 + 2.5)}{512 - 8} \right]$$

$$= 1 - \left[\frac{(6 \times 94)}{504} \right]$$

$$= 1 - \frac{564}{504}$$

$$= 1 - 1.1190$$

$$r = -0.119$$

there is lower degree negative correlation plus & 87.

~~20~~ correct

Regression

Find unknown value.

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Regression is the theory of estimation of unknown values of variable with a help of known values of variable

For a bivariate data on X and Y, the regression equation obtained with a assumption that X is dependent on Y is called regression of X on Y, that is

$$(X - \bar{X}) = b_{YX} (Y - \bar{Y})$$

Y is unknown
X is known

The regression equation obtained with a assumption that Y is dependent on X is called Regression of Y on X, that is

$$(Y - \bar{Y}) = b_{XY} (X - \bar{X})$$

Y is unknown
X is known

the b_{YX} and b_{XY} are regression coefficients they are

$$b_{YX} = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (Y - \bar{Y})^2}$$

OR

$$b_{YX} = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum Y^2 - (\sum Y)^2}$$

} b_{YX}

$$b_{YX} = r \frac{\sigma_Y}{\sigma_X}$$

OR

$$b_{YX} = \frac{n \sum XY - (\sum X)(\sum Y)}{n \sum X^2 - (\sum X)^2}$$

} b_{YX}

PROPERTIES

Properties of Regression coefficients.

① The geometric mean of the regression coefficients is equal to coefficient of correlation relation b/c

$$① \quad r = \pm \sqrt{b_{YX} \cdot b_{XY}}$$

← correlation relation b/c

② The regression coefficients can not be of opposite signs.

- If r is +ve, the regression coefficients will be positive
- If r is -ve, both regression coefficients will be negative
- If $r = 0$, both regression coefficients will be zero.

③ The product of regression coefficient can not be greater than one.

$$b_{YX} \times b_{XY} < 1$$

$$\frac{25-10}{10} \times \frac{21-10}{10} = \frac{15}{10} \times \frac{11}{10} = \frac{165}{100} = 1.65 > 1$$

$$(10) \quad \frac{21-10}{10} \times \frac{21-10}{10} = \frac{11}{10} \times \frac{11}{10} = \frac{121}{100} = 1.21 > 1$$

$$\frac{25-10}{10} \times \frac{21-10}{10} = \frac{15}{10} \times \frac{11}{10} = \frac{165}{100} = 1.65 > 1$$

$$\frac{25-10}{10} \times \frac{21-10}{10} = \frac{15}{10} \times \frac{11}{10} = \frac{165}{100} = 1.65 > 1$$

and according to above two properties, we can say that the regression coefficients are not correct.

Q1. Find the regression equation of Y on X for the following data.

- Q. A survey of children within the following population regarding the age of child & age of mother and the type of giving birth to child

	x	y
Woman	98	38 years
Girl	82	40 years

coefficient of correlation = -0.84

Estimate age of child whose mother was aged 44 at the time of giving birth to the child.

$$\bar{x} = 90 \quad \bar{y} = 39 \quad r = -0.84 \quad \text{when } y = 44 \text{ years}$$

$$x \text{ on } y$$

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$b_{yx} = \frac{\sum xy}{\sum x^2}$$

$$= b_{yx} = -0.84 \times \frac{1}{1}$$

$$b_{yx} = -0.12$$

$$x - 98 = -0.12 (44 - 38)$$

$$x - 98 = -0.12 (6)$$

$$x - 98 = -0.72$$

$$x = -0.72 + 98$$

$$x = 97.28$$

- Q. The following are heights of 8 persons and 1 each of their sons from the data estimate with height of person whose father is 150cm tall.

Height of father	164	176	178	187	175	167	172	180
Height of son	168	174	175	181	173	166	173	179

x	y	x ²	xy
164	168	26896	27552
176	174	30976	30624
178	175	31684	31150
187	181	34969	33857
175	173	30625	30275
167	166	27889	27722
173	173	29929	29929
180	179	32400	32220
179	173	32041	30963

$$y \text{ on } x$$

$$\bar{y} = \frac{\sum y}{n}$$

$$b_{yx} = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{8 \times 244855 - (1397) \times 1954090}{8 \times 244855 - (1397)^2}$$

$$= \frac{1954090 - 1951609}{2431}$$

$$= 0.7301$$

$$= 1745$$

$$= 1745$$

$$= 1745$$

$$= 1745$$

$$= 1745$$

$$= 1745$$

$$= 1745$$

$$= 1745$$

$$= 1745$$

$$= 1745$$

$$= 1745$$

y on x.

$$y - \bar{y} = byx (x - \bar{x})$$

$$y - 173.685 = 0.7301 (150 - 174.685)$$

$$y - 173.685 = 0.7301 (-24.685)$$

$$y - 173.685 = -17.9787$$

$$y = -17.9787 + 173.685$$

$$y = 155.6463$$

③ the height (h) in cms weight in kgs of 6 students

we can follow the method to find the expected height of person whose weight 60 kgs.

x	y	x ²	y ²	xy
153	48	23409	2304	7344
154	50	23649	2500	7850
168	50	28224	2500	8400
160	49	25600	2401	7840
170	54	28900	2916	9180
163	53	26569	2809	8630
971	304	157351	15430	49853

2 reg. u.

n=6

① $\frac{\sum xy}{\sum x} = byx (y - \bar{y})$

$$\bar{x} = \frac{\sum x}{n} = \frac{971}{6} = 161.833$$

$$\bar{y} = \frac{\sum y}{n} = \frac{304}{6} = 50.666$$

$$byx = \frac{\sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$n \sum y^2 - (\sum y)^2$$

$$byx = \frac{6 \times 49853 - (971) \times (304)}{6 \times 157351 - (971)^2}$$

$$= \frac{6 \times 49853 - 294116}{6 \times 157351 - 942881}$$

$$= \frac{299118 - 294116}{944106 - 942881}$$

$$= \frac{5002}{2225}$$

$$= \frac{2.247}{1} = 2.247$$

$$= \frac{299118 - 294116}{944106 - 942881}$$

$$= \frac{5002}{2225} = 2.247$$

$$byx = 2.0365$$

$$(x - \bar{x}) = byx (y - \bar{y})$$

$$(x - 161.833) = 2.0365 (y - 50.666)$$

$$x - 161.833 = 2.0365y - 103.181 + 161.833$$

$$x - 161.833 = 2.0365y - 103.181$$

$$x = 2.0365y + 58.652$$

$$x = 2.0365y + 58.652$$